# Generating Indistinguishable Single-Photons from Strongly Dissipative Quantum Emitters using Cavity-Funneling in GaN

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# 1 Introduction

Indistinguishable single photons are an essential resource for a number of optical quantum information protocols, from linear optical quantum computing and boson sampling to quantum teleportation and quantum networks. Two photons are said to be indistinguishable when no measurement can tell them apart, so that they interfere on a beam splitter as predicted by the Hong-Ou-Mandel effect. In practice this requires the emitted photons to share the same spectrum, polarisation and temporal envelope (the photons have to be so-called Fourier-transform limited).
A natural way to produce such photons on demand is to use a single two-level quantum emitter such as an atom, a colour centre, a quantum dot or an organic molecule. This works very well at cryogenic temperatures, but as soon as the emitter is operated at higher temperatures pure dephasing becomes the dominant decoherence mechanism. Phonon-induced dephasing broadens the emitter's homogeneous linewidth far beyond its radiative limit, so successive photons are no longer Fourier-transform limited and their indistinguishability collapses. For most solid-state emitters at room temperature the pure dephasing rate $\gamma^*$ exceeds the radiative rate $\gamma$ by three to six orders of magnitude, and the intrinsic indistinguishability is essentially zero.
Two strategies are commonly used to recover indistinguishability. The first is to spectrally filter the emission, keeping only a narrow window of the broad emitted spectrum. This restores Fourier-transform-limited photons but at the cost of a vanishingly small efficiency, since most of the emitted light is discarded. The second is to place the emitter inside a cavity and rely on the Purcell effect to enhance the radiative rate above the dephasing rate. For room-temperature solid-state systems this requires Purcell factors larger than $\gamma^*/\gamma$, which is well beyond the current state of the art.
A more recent proposal by Grange et al. [1] circumvents both limitations by exploiting an unconventional cavity-QED regime, in which a high-quality-factor cavity with moderate emitter-cavity coupling acts as a spectral funnel: the broad emission of the dissipative emitter is redirected into the narrow cavity line. The resulting photons inherit the cavity's coherence properties rather than the emitter's, and the predicted efficiency-indistinguishability product can exceed any spectral-filtering scheme by orders of magnitude. Because this regime relies on cavity parameters that are within reach of current nanofabrication, it opens a realistic path towards on-chip indistinguishable single-photon sources operating at room temperature.
This semester project investigates whether this strategy can be applied to gallium nitride (GaN) emitter-cavity systems, the material platform of the host laboratory. The work proceeds in three steps. First, we recover and re-derive the efficiency $\beta$ and the indistinguishability $I$ predicted by Grange's model. Second, we use these results to evaluate a selection of state-of-the-art GaN cavities reported in the literature, and locate them on the $(g, \kappa)$ map to assess how close current GaN platforms are to the funneling regime. Finally, we discuss what cavity geometry and parameters would be needed to push GaN into this regime, and what an experimentally realistic device might look like.

# 2 Theory

Before getting into the derivations, it is worth clarifying the theoretical framework we work in.

## 2.1 Efficiency and indistinguishability

Throughout this report, the performance of an emitter-cavity source is characterised by two dimensionless numbers between 0 and 1.
The efficiency $\beta$ is the probability that a single excitation of the emitter results in a photon collected through the cavity output, rather than lost to the environment:

$$\beta = \kappa \int_0^\infty \langle \hat{n}_{\text{cav}}(t) \rangle \, dt$$

where $\hat{n}_{\text{cav}} = |g,1\rangle\langle g,1|$ is the cavity photon-number operator. An ideal source has $\beta = 1$; spectral filtering schemes typically have $\beta \ll 1$.
The indistinguishability $I$ is the visibility of the Hong-Ou-Mandel dip when two successively emitted photons interfere on a 50/50 beam splitter. As we will see, it can be written in terms of the cavity field correlators as :

$$I = \frac{\displaystyle\int_0^\infty dt \int_0^\infty d\tau \left|\langle \hat{a}^\dagger(t+\tau)\hat{a}(t)\rangle\right|^2}{\displaystyle\int_0^\infty dt \int_0^\infty d\tau \, \langle \hat{a}^\dagger(t)\hat{a}(t)\rangle \, \langle \hat{a}^\dagger(t+\tau)\hat{a}(t+\tau)\rangle}$$

$I = 1$ corresponds to perfectly identical (Fourier-transform-limited) photons and $I = 0$ to fully distinguishable ones.

## 2.2 The Lindblad master equation

A closed quantum system evolves unitarily under the Schrödinger equation, fully described by its state vector $|\psi(t)\rangle$. Our system, however, is open: the emitter and the cavity continuously exchange energy and coherence with their surroundings (the electromagnetic vacuum, etc. . . ). A pure state vector cannot represent the resulting statistical mixture, so we describe the system by its density matrix $\rho$. Under the Markov approximation, the evolution of $\rho$ is governed by the Lindblad master equation, the natural generalisation of the Schrödinger equation to open systems.
We work in the single-excitation Hilbert space spanned by $\{|g,0\rangle, |g,1\rangle, |e,0\rangle\}$, where the first label denotes the emitter state (ground $|g\rangle$ or excited $|e\rangle$) and the second the cavity Fock state (0 or 1 photon). Two-excitation states such as $|e,1\rangle$ are excluded by the rotating-wave approximation combined with the choice of an instantaneous single excitation as initial condition: the system never contains more than one quantum. We further set the emitter-cavity detuning to zero, that is, in perfect resonance.
The density matrix evolves under the Lindblad master equation :

$$\frac{d\rho}{dt} = \mathcal{L}[\rho] = -\frac{i}{\hbar}[H,\rho] + \sum_k \left( L_k \rho L_k^\dagger - \frac{1}{2}\left\{ L_k^\dagger L_k ,\, \rho \right\} \right) \tag{1}$$

The first term is the usual closed-system part, generated by the Hamiltonian

$$H = \hbar g(|e,0\rangle\langle g,1| + |g,1\rangle\langle e,0|) \tag{2}$$

which describes the coherent exchange of one excitation between the emitter and the cavity at the dipolar coupling rate $g$. The second term encodes the irreversible coupling to the environment through three collapse operators $L_k$, each associated with one dissipative channel:

- $L_1 = \sqrt{\gamma}\ |g,0\rangle\langle e,0|$: spontaneous emission of the emitter into non-cavity modes at rate $\gamma$. These photons are lost, bad channel.
- $L_2 = \sqrt{\kappa}\ |g,0\rangle\langle g,1|$: leakage of the cavity photon into the outgoing mode at rate $\kappa$. This is the desired output, good channel.
- $L_3 = \sqrt{\gamma^*}\ |e,0\rangle\langle e,0|$: pure dephasing at rate $\gamma^*$, which randomises the phase of the emitter coherence without transferring population. It is the dominant decoherence mechanism at room temperature.

We initialise the system with one excitation in the emitter, $\rho_0 = |e,0\rangle\langle e,0|$, and follow its evolution under (1) until all population has decayed.
Figure 1 below shows the a schematic of our system.

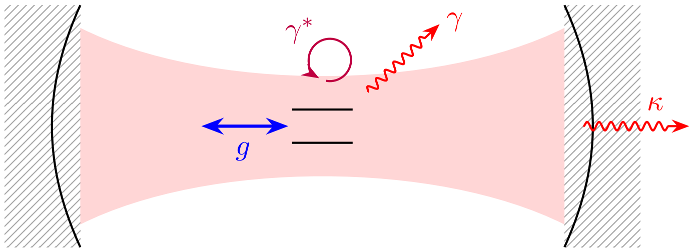



Figure 1: Schematic of a dissipative two-level emitter coupled to an optical cavity mode.

### 2.3 Approximations

The model written above is compact, but it rests on many approximations, some of which we can discuss here. This list is likely not exhaustive, but it just serves to highlight the fact that the validity of the model is somewhat limited, and wouldn't imply that realising such a system would necessarily be possible.
The first one is the Markov approximation, on which the Lindblad form of equation (1) relies. It assumes that the environment has no memory: the bath correlation time is much shorter than any system timescale ($1/g$, $1/\kappa$, $1/\gamma^*$). For solid-state emitters at room temperature, the phonon bath relaxes on a picosecond timescale, comfortably faster than the dynamics we consider here, so this is well justified.
A second one is the rotating-wave approximation (RWA), in which we drop the counter-rotating terms in the emitter-cavity coupling. This requires $g \ll \omega_0$, where $\omega_0$ is the optical frequency. For the cavities relevant to this work, $\omega_0 \sim 10^{15}$ rad/s and the coupling rates considered are at most $g \sim 10^{12}$ rad/s, this holds by at least three orders of magnitude.
The emitter and cavity are also highly simplified. They are restricted to the single-excitation subspace, and the emitter is described as a strict a two level system. The cavity is considered to have a singly mode, and all the rates of the system are considered constant in time. These approximations are more difficult to quantify, but are likely strong assumptions.
Finally, we assume zero detuning, meaning that the emitter is perfectly resonant with the cavity mode ($\delta = 0$). This is convenient and gives the best-case figures of merit, but it is an idealisation, as will we see later in Section 6.

With this model and its limitations in place, we can now start the derivation of $\beta$ and $I$ from the Lindblad equation (1).

## 3 Efficiency

The efficiency $\beta$ is defined as the probability that a photon, following the initial excitation of the emitter, is emitted through the cavity's good channel rather than lost to the environment. Formally, it is the total flux of photons leaving through the cavity mode:

$$\beta = \kappa \int_0^\infty \langle \hat{n}_{\text{cav}}(t)\rangle \, dt = \kappa \int_0^\infty \text{Tr}[P_{\text{cav}}\, \rho(t)] \, dt \tag{3}$$

where $P_{\text{cav}} = |g,1\rangle\langle g,1|$ is the projector onto the cavity-occupied state and $\kappa$ is the cavity decay rate. Physically, this makes a lot of sense as it is intuitively the average number of photons in the cavity over time, weighted by $\kappa$, the rate at which each photon escapes through the good channel.

### 3.1 Numerical approach

The first attempt to obtain $\beta$ was done by naively integrating the Lindblad equation (1) numerically and evaluate the integral (3) by quadrature on a $(g, \kappa)$ grid. We used QuTiP's master-equation solver `mesolve` in the three-state basis $\{|g,0\rangle, |g,1\rangle, |e,0\rangle\}$, with the three collapse operators $L_1$, $L_2$, $L_3$ and the initial condition $\rho_0 = |e,0\rangle\langle e,0|$. The expectation value $\langle \hat{n}_{\mathrm{cav}}(t)\rangle$ was returned on a uniform time grid and integrated with the trapezoidal rule. For each $(g, \kappa)$ point, the computation requires two separate convergence studies:

- the number of time steps $n_t$ must be large enough to resolve the fastest dynamics in the system (the rates $g$, $\kappa$, $\gamma^*$);
- the time horizon $t_{\max}$ must be long enough that essentially all the population has decayed, so that the integral (3) has converged.

To ensure the first convergence, we ran multiple simulations at fixed $t_{\max}$ while doubling $n_t$ until the change in the final value $\beta(t_{\max})$ was smaller than a set tolerance.
For the second, we used a fixed time step, but increase the final time $t_{\max}$ to detect when the value of $\beta(t_{\max})$ plateaued (i.e, two consecutive values has a relative change smaller than the tolerance), doubling $t_{\max}$ otherwise. A typical convergence study is shown in Fig. 2.

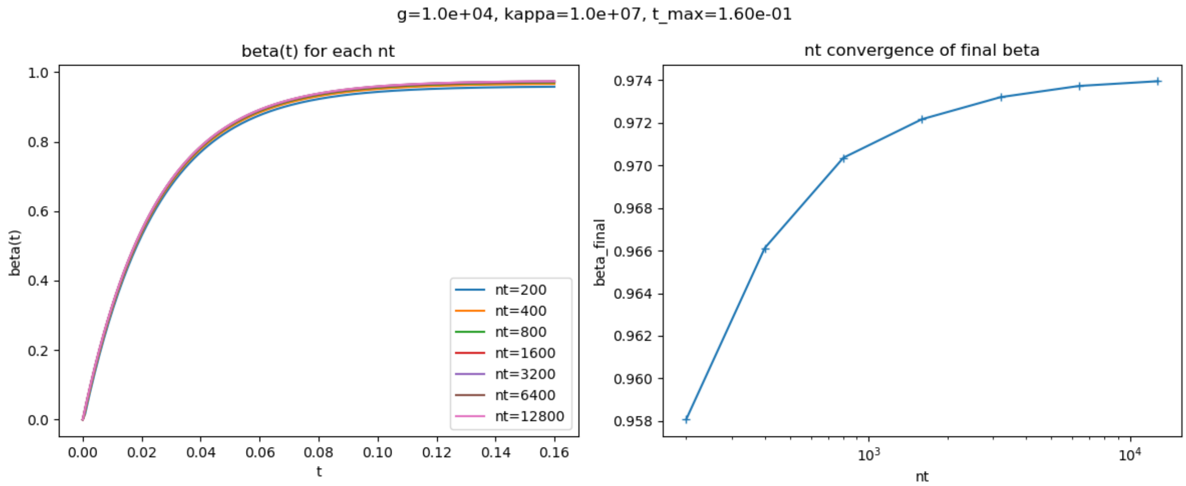



Figure 2: Convergence of $\beta(t)$ at the representative point $(g, \kappa) = (10^4, 10^7)\,\gamma$. Left: $\beta(t)$ for several values of $n_t$. Right: final value $\beta(t_{\max})$ as a function of $t_{\max}$.

This method worked well for bottom left region of the phase space. However, as soon as we attempted to run the simulations on the whole space, we quickly ran into exponentially growing computation times.
At first, several refinements were attempted to push the solver into the harder regions of the parameter space: the stiff backward-differentiation formula (BDF) integrator, a large internal step budget ($n_{\mathrm{steps}} = 2{\times}10^6$), considerably looser tolerances, an adaptative $t_{\max}$ routine doubling the horizon until $\beta(t)$ converged, and a non-uniform time mesh concentrating simulation points where $\beta(t)$ varies fastest, in order to minimize the number of simulations to run. Computations were also parallelised to increase the computation speed sereval folds. Despite all of this, the solver still failed to return a converged value in a significant fraction of the $(g, \kappa)$ plane.

#### 3.1.1 Failure of the numerical approach

After spending a considerable amount of time spent trying to eliminate sources of complexity, it was clear the cause of the failure of the numerical approach was "internal" to the problem.

The root cause was stiffness. The evolution of the system deals with timescales that differ by many orders of magnitude. Looking at the physical meaning of the parameters give us a strong intuition of the problem. First, recall that we work with normalised parameters $\kappa/\gamma$ and $g/\gamma$. They respectively give the decay rate of the cavity and the coherent exchange rate of the QE-cavity pair, relative to the emission rate of the QE.
This intuitively means that for $\kappa/\gamma < 1$ and $g/\gamma < 1$, the largest rate to consider (i.e, the component of the system with the fastest time evolution) is the emission from the QE to the cavity. Howerever, when either $\kappa \ll \gamma$ or $g \ll \gamma$, the time evolution has to account for tiny time steps, where the cavity population could decay, or the Rabi oscillations can occur, respectively. The integrator must therefore simultaneously resolve the fast processes and run up to a horizon set by the slow ones. In any region of the phase space other than the bottom left corner, this quickly grows by orders of magnitude.
This is a structural limitation rather than a tuning problem: no choice of tolerance, step count or grid spacing can fix it without making the cost grow faster than the parameter range of interest. This observation motivated the search for an analytical expression for $\beta$, which is the subject of the next section.

## 3.2 Analytical approach

### 3.2.1 Vectorisation of the density matrix

The Lindblad equation is already linear in $\rho$, but in its native form it acts on a matrix rather than a vector. Vectorisation rewrites the same linear map as an ordinary matrix-vector product, by stacking the columns of $\rho$ into a single column vector:

$$\rho \longmapsto |\rho\rangle\rangle \in \mathbb{C}^{d^2} \tag{4}$$

In our case, the Hilbert space $\{|g,0\rangle, |g,1\rangle, |e,0\rangle\}$ has dimension $d = 3$, so $\rho$ is a $3 \times 3$ matrix with $d^2 = 9$ complex entries, and vectorisation maps it to a column vector $|\rho\rangle\rangle \in \mathbb{C}^9$. The Liouvillian, which previously acted as a superoperator on $3 \times 3$ matrices, is then represented as an ordinary $9 \times 9$ matrix $\hat{\mathcal{L}}$ acting on $|\rho\rangle\rangle$.
This turns the Lindblad equation into an ordinary linear ODE:

$$\frac{d}{dt}|\rho\rangle\rangle = \hat{\mathcal{L}}\,|\rho\rangle\rangle \implies |\rho(t)\rangle\rangle = e^{\hat{\mathcal{L}}t}|\rho_0\rangle\rangle \tag{5}$$

$\hat{\mathcal{L}}$ is a $9 \times 9$ complex matrix called the Liouvillian superoperator. Using the identity $|AXB\rangle\rangle = (B^\top \otimes A)\,|X\rangle\rangle$, it takes the explicit form:

$$\hat{\mathcal{L}} = -\frac{i}{\hbar}(I \otimes H - H^\top \otimes I) + \sum_k \left[L_k^* \otimes L_k - \frac{1}{2}\Big(I \otimes L_k^\dagger L_k + (L_k^\dagger L_k)^\top \otimes I\Big)\right] \tag{6}$$

Expectation values then become inner products in the vectorised space:

$$\mathrm{Tr}[P_{\mathrm{cav}}\,\rho(t)] = \langle\langle P_{\mathrm{cav}}|\,\rho(t)\rangle\rangle \tag{7}$$

so that the efficiency (3) reads:

$$\beta = \kappa \int_0^\infty \langle\langle P_{\mathrm{cav}}|\,e^{\hat{\mathcal{L}}t}|\rho_0\rangle\rangle\,dt \tag{8}$$

### 3.2.2 The singular Liouvillian

All eigenvalues of $\hat{\mathcal{L}}$ have non-positive real parts, and there is a unique zero eigenvalue corresponding to the steady state $\rho_{\mathrm{ss}} = |g,0\rangle\langle g,0|$ (all excitation has decayed). This makes $\hat{\mathcal{L}}$ singular,

so one cannot directly invert the integral in (8).
The standard workaround is to split $\rho(t)$ into its asymptotic part and a decaying remainder:

$$\rho(t) = \rho_{\rm ss} + \sigma(t), \qquad \sigma(t) \equiv \rho(t) - \rho_{\rm ss} \tag{9}$$

By construction $\sigma(t)$ contains only the components of $\rho$ that decay in time, i.e. the projection of $\rho$ onto the non-null modes of $\hat{\mathcal{L}}$. Working with $\sigma$ instead of $\rho$ confines the problem to the subspace where $\hat{\mathcal{L}}$ is invertible.
Two properties of $\sigma$ will be useful. First, $\sigma$ satisfies the same Lindblad equation as $\rho$: since $\hat{\mathcal{L}}$ is linear and $\hat{\mathcal{L}}\,|\rho_{\rm ss}\rangle\!\rangle = 0$,

$$\frac{d}{dt}\,|\sigma\rangle\!\rangle = \frac{d}{dt}\,|\rho\rangle\!\rangle = \hat{\mathcal{L}}\,|\rho\rangle\!\rangle = \hat{\mathcal{L}}\,|\sigma\rangle\!\rangle$$

Second, $\sigma(\infty) = 0$ because $\rho(t) \to \rho_{\rm ss}$ as $t \to \infty$. This is what allows the boundary term below to collapse cleanly. Integrating $\frac{d}{dt}\,|\sigma\rangle\!\rangle = \hat{\mathcal{L}}\,|\sigma\rangle\!\rangle$ from 0 to $\infty$:

$$\hat{\mathcal{L}} \int_0^\infty |\sigma(t)\rangle\!\rangle\, dt = \left[ e^{\hat{\mathcal{L}}t}\,|\sigma_0\rangle\!\rangle \right]_0^\infty = -\,|\sigma_0\rangle\!\rangle \tag{10}$$

where $|\sigma_0\rangle\!\rangle = |\rho_0 - \rho_{\rm ss}\rangle\!\rangle$.
At this point $\hat{\mathcal{L}}$ is still singular as a $9 \times 9$ matrix, but $|\sigma_0\rangle\!\rangle$ lives entirely in its range, where the inversion is well-defined. The notation $\hat{\mathcal{L}}^{-1}$ below therefore should be read as the pseudo-inverse which acts as a genuine inverse on $|\sigma_0\rangle\!\rangle$ since $|\sigma_0\rangle\!\rangle$ lies in that subspace:

$$\int_0^\infty |\sigma(t)\rangle\!\rangle\, dt = -\hat{\mathcal{L}}^{-1}\,|\sigma_0\rangle\!\rangle \tag{11}$$

Since $\mathrm{Tr}[P_{\rm cav}\,\rho_{\rm ss}] = 0$ (there are no photons in the cavity at steady state), we have

$$\int_0^\infty \langle\!\langle P_{\rm cav}|\,\rho(t)\rangle\!\rangle\, dt = \int_0^\infty \langle\!\langle P_{\rm cav}|\,\sigma(t)\rangle\!\rangle\, dt \tag{12}$$

and the efficiency reduces to the closed-form expression:

$$\boxed{\beta = -\kappa\,\langle\!\langle P_{\rm cav}|\,\hat{\mathcal{L}}^{-1}|\rho_0 - \rho_{\rm ss}\rangle\!\rangle} \tag{13}$$

In practice this reduces to a single $9 \times 9$ linear solve $\hat{\mathcal{L}}\,x = -\,|\rho_0 - \rho_{\rm ss}\rangle\!\rangle$, which is exact and can be computed numerically very fast, in a time independent of the values of $g/\gamma$, $\kappa/\gamma$, or $\gamma^*/\gamma$.

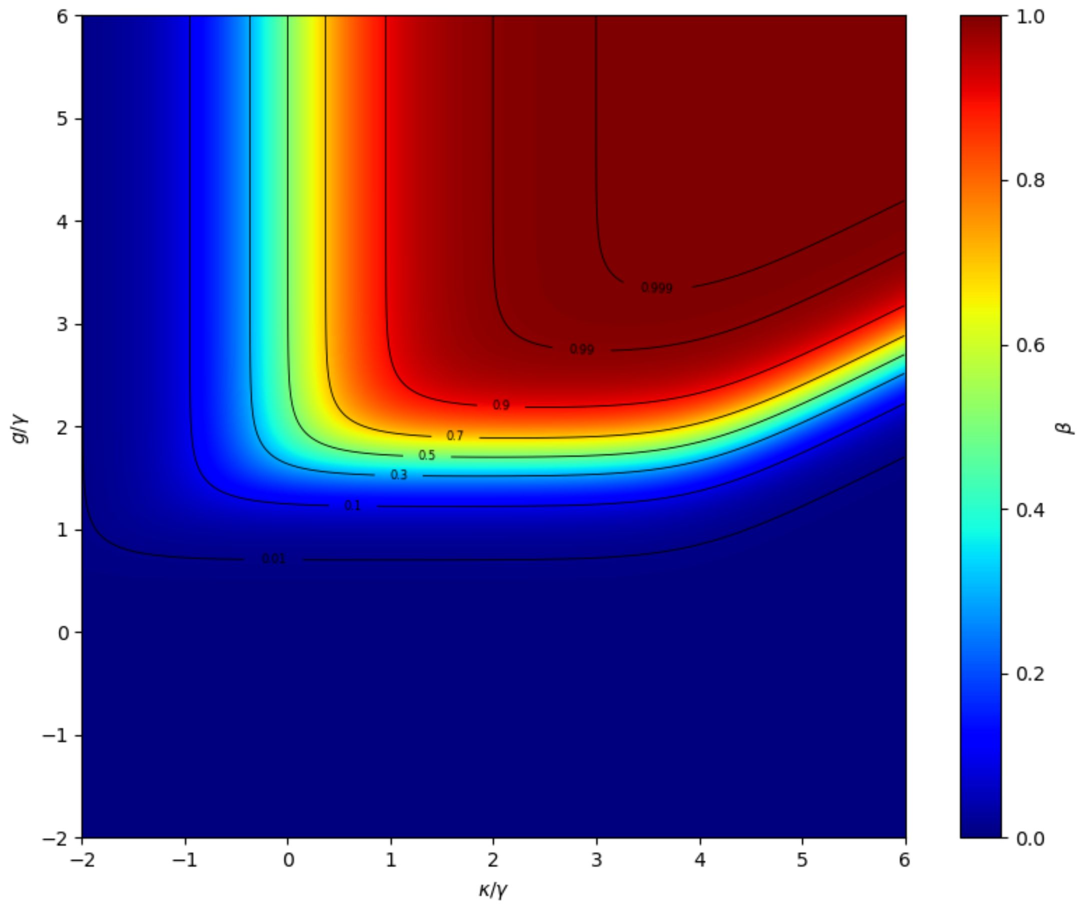



Figure 3: Efficiency $\beta$ as a function of the coupling $g$ and cavity decay rate $\kappa$, for a fixed ratio $\gamma^*/\gamma = 10^4$.

### 3.3 Alternative analytical approach

In hindsight, it was possible to solve this more easily, without the vectorisation. Howerever, the formalism is useful for the part on indistinguishability so we accept to pay that price, and it is already done anyways.
Still, we can sketch the approach, as a system of ODE.
In the basis $\{|g,0\rangle, |g,1\rangle, |e,0\rangle\}$, $\rho$ is:

$$\rho = \begin{pmatrix} \rho_{gg} & \rho_{gc} & \rho_{ge} \\ \rho_{cg} & \rho_{cc} & \rho_{ce} \\ \rho_{eg} & \rho_{ec} & \rho_{ee} \end{pmatrix} \tag{14}$$

Each element evolves as $\dot{\rho}_{ij} = \langle i| \dot{\rho} |j\rangle$, given by the Lindblad equation (1), and we can evaluate it.

#### 3.3.1 First term

The commutator $[H, \rho]$ gives for each element:

$$\langle i| [H, \rho] |j\rangle = \langle i| H\rho |j\rangle - \langle i| \rho H |j\rangle \tag{15}$$

with, as a reminder, the Hamiltonian $H = \hbar g(|e,0\rangle\langle g,1| + |g,1\rangle\langle e,0|)$.
For example for $\dot{\rho}_{ee}$:

$$
\begin{aligned}
-\frac{i}{\hbar}\langle e,0|\,[H,\rho]\,|e,0\rangle &= -\frac{i}{\hbar}\Big(\langle e,0|\,H\rho\,|e,0\rangle - \langle e,0|\,\rho H\,|e,0\rangle\Big) \\
&= -\frac{i}{\hbar}\left(\sum_k \langle e,0|\,H\,|k\rangle\,\langle k|\,\rho\,|e,0\rangle - \sum_l \langle e,0|\,\rho\,|l\rangle\,\langle l|\,H\,|e,0\rangle\right) \\
&= -ig\left(\rho_{ce} - \rho_{ec}\right)
\end{aligned}
$$

The same can be done for each element of the matrix. Note however, than the Hamiltonian only has elements in $|g,1\rangle, |e,0\rangle$. Therefore, it acts trivially on $|g,0\rangle$ (all matrix elements involving $|g,0\rangle$ vanish), and it commutes with the number operator $\hat{N}$ so the subspaces with fixed excitation number are invariant.
After some simple computation that can be found in appendix A, we finally get :

$$
\begin{aligned}
-\frac{i}{\hbar}[H,\rho]_{ee} &= -ig(\rho_{ce} - \rho_{ec}) \\
-\frac{i}{\hbar}[H,\rho]_{cc} &= +ig(\rho_{ce} - \rho_{ec}) \\
-\frac{i}{\hbar}[H,\rho]_{ec} &= +ig(\rho_{ee} - \rho_{cc})
\end{aligned}
$$

### 3.3.2 Second term

The contribution of each operator $L_k$ is:

$$L_k\rho L_k^\dagger - \frac{1}{2}(L_k^\dagger L_k\rho + \rho L_k^\dagger L_k)$$

Projected onto element $(i,j)$:

$$\langle i|\,L_k\rho L_k^\dagger\,|j\rangle - \frac{1}{2}(\langle i|\,L_k^\dagger L_k\rho\,|j\rangle + \langle i|\,\rho L_k^\dagger L_k\,|j\rangle)$$

Take $L_1 = \sqrt{\gamma}\,|g,0\rangle\langle e,0|$ as an example.
First, $L_1^\dagger L_1 = \gamma\,|e,0\rangle\langle e,0| = \gamma P_e$, so the anticommutator term gives:

$$-\frac{\gamma}{2}\langle i|\,\{P_e,\rho\}\,|j\rangle = -\frac{\gamma}{2}(\langle i|\,P_e\rho\,|j\rangle + \langle i|\,\rho P_e\,|j\rangle)$$

And the $L_1\rho L_1^\dagger$ term gives:

$$\gamma\,\langle i|g,0|i|g,0\rangle\,\langle e,0|\,\rho\,|e,0\rangle\,\langle g,0|j|g,0|j\rangle = \gamma\,\delta_{i,g_0}\delta_{j,g_0}\,\rho_{ee}$$

So $L_1$ drains $\rho_{ee}$ into $\rho_{gg}$ at rate $\gamma$, and kills all coherences involving $|e,0\rangle$ at rate $\gamma/2$.
The same procedure is applied to $L_2 = \sqrt{\kappa}\,|g,0\rangle\langle g,1|$ and $L_3 = \sqrt{\gamma^*}\,|e,0\rangle\langle e,0|$.

### 3.3.3 Combined terms

Now, applying the dissipator contributions to each equations yields:

$$
\begin{aligned}
\dot{\rho}_{ee} &= -\gamma\,\rho_{ee} - ig(\rho_{ce} - \rho_{ec}) \\
\dot{\rho}_{cc} &= -\kappa\,\rho_{cc} + ig(\rho_{ce} - \rho_{ec}) \\
\dot{\rho}_{ec} &= -\frac{\gamma+\kappa+\gamma^*}{2}\,\rho_{ec} + ig(\rho_{ee} - \rho_{cc})
\end{aligned}
$$

By hermiticity, $\rho_{ce} = \rho_{ec}^*$, so $\rho_{ce} - \rho_{ec} = -2i\,\mathrm{Im}(\rho_{ec})$. Defining $v = \mathrm{Im}(\rho_{ec})$, the system simplifies to three real equations:

$$\boxed{\begin{aligned} \dot\rho_{ee} &= -\gamma\,\rho_{ee} - 2g\,v \\ \dot\rho_{cc} &= -\kappa\,\rho_{cc} + 2g\,v \\ \dot v &= g(\rho_{ee} - \rho_{cc}) - \frac{\gamma+\kappa+\gamma^*}{2}\,v \end{aligned}} \tag{16}$$

This is $\dot{\mathbf{x}} = M\mathbf{x}$ with $\mathbf{x} = (\rho_{ee}, \rho_{cc}, v)^\top$ and

$$M = \begin{pmatrix} -\gamma & 0 & -2g \\ 0 & -\kappa & 2g \\ g & -g & -\dfrac{\gamma+\kappa+\gamma^*}{2} \end{pmatrix}$$

and initial condition $\mathbf{x}(0) = (1,0,0)^\top$. To get $\beta$, define $\mathbf{X} = \int_0^\infty \mathbf{x}(t)\,dt$. Integrating $\dot{\mathbf{x}} = M\mathbf{x}$ directly gives $M\mathbf{X} = -\mathbf{x}(0)$, since $\mathbf{x}(\infty) = 0$. $M$ is invertible because $|g,0\rangle$ is not tracked here. More precisely, $\rho_{gg}$ is fixed by trace conservation ($\rho_{gg} = 1 - \rho_{ee} - \rho_{cc}$, similarly to the singular Liouvillian) so the reduced dynamics only has decaying modes, and no zero eigenvalue.
Solving this system gives is simple numerically, we get $X_{cc} = \int_0^\infty \rho_{cc}(t)\,dt$ and

$$\beta = \kappa\,X_{cc} \tag{17}$$

As a sanity check, Fig. 4 shows the efficiency computed this way, and it is identical to the result obtained via the previous method. The computation is moreover cheaper, operating in a $3 \times 3$ rather than $9 \times 9$ space.

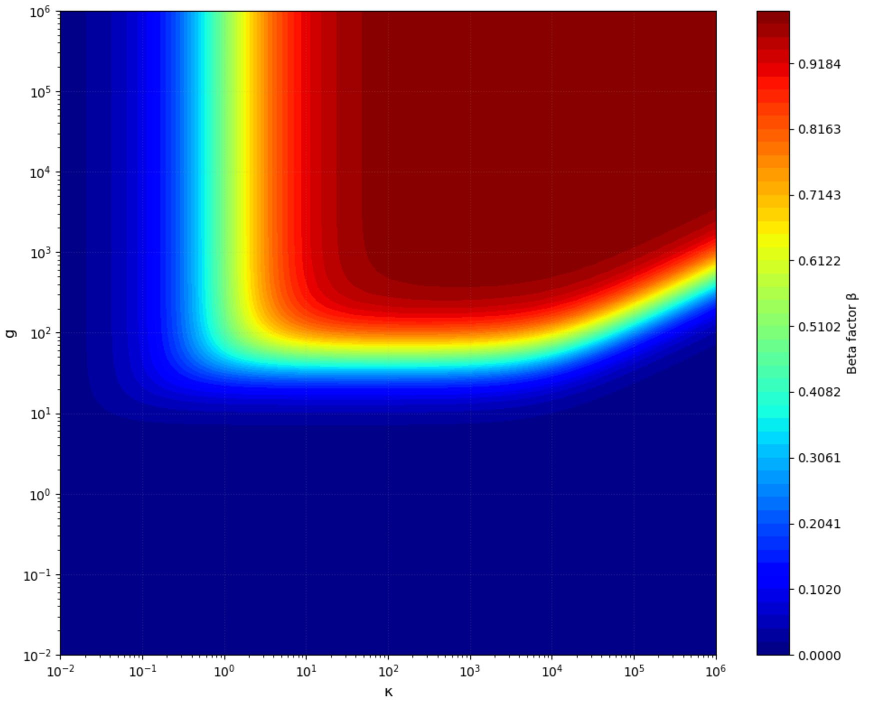



Figure 4: Efficiency $\beta$ computed via the reduced $3 \times 3$ system.

## 4 Indistinguishability

Having computed the efficiency analytically, we now turn to the indistinguishability.

## 4.1 Hong-Ou-Mandel indistinguishability

For a single-photon emitter to be useful in quantum information applications, it is not enough that it emits one photon at a time: successive photons must also be identical, i.e. indistinguishable. Indistinguishability quantifies how well two photons from the same source can interfere, measured by the Hong-Ou-Mandel experiment.

### 4.1.1 The Hong-Ou-Mandel effect

The Hong-Ou-Mandel (HOM) experiment probes the quantum indistinguishability of two single photons by sending them onto a 50/50 beam splitter (BS) and monitoring coincidences at the two output ports.
The setup is shown in Figure 5: two photons enter the BS through input modes $\hat{a}_1$ and $\hat{a}_2$, and exit through output modes $\hat{a}_3$ and $\hat{a}_4$. For a symmetric 50/50 BS, the input and output modes are related by:

$$\hat{a}_3 = \frac{1}{\sqrt{2}}(\hat{a}_1 + \hat{a}_2),\ \hat{a}_4 = \frac{1}{\sqrt{2}}(\hat{a}_1 - \hat{a}_2) \tag{18}$$

Inverting these relations gives $\hat{a}_1^\dagger = (\hat{a}_3^\dagger + \hat{a}_4^\dagger)/\sqrt{2}$ and $\hat{a}_2^\dagger = (\hat{a}_3^\dagger - \hat{a}_4^\dagger)/\sqrt{2}$. The two-photon input state $|1,1\rangle \equiv \hat{a}_1^\dagger \hat{a}_2^\dagger |0,0\rangle$ is therefore mapped to:

$$\begin{aligned} \hat{a}_1^\dagger \hat{a}_2^\dagger |0,0\rangle &= \frac{1}{2}(\hat{a}_3^\dagger + \hat{a}_4^\dagger)(\hat{a}_3^\dagger - \hat{a}_4^\dagger) |0,0\rangle \\ &= \frac{1}{2}(\hat{a}_3^{\dagger 2} - \hat{a}_3^\dagger \hat{a}_4^\dagger + \hat{a}_4^\dagger \hat{a}_3^\dagger - \hat{a}_4^{\dagger 2}) |0,0\rangle \end{aligned} \tag{19}$$

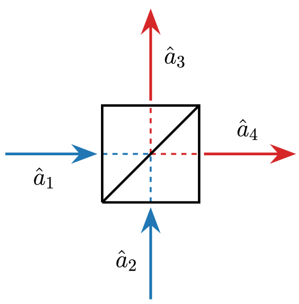



Figure 5: Schematic of the HOM experiment.

The four terms correspond to the four possible scattering outcomes, illustrated in Figure 6:

(1) $\hat{a}_1$ reflected, $\hat{a}_2$ transmitted $\to$ both exit in mode $\hat{a}_3$

(2) both photons transmitted $\to$ one photon in each output

(3) both photons reflected $\to$ one photon in each output

(4) $\hat{a}_1$ transmitted, $\hat{a}_2$ reflected $\to$ both exit in mode $\hat{a}_4$

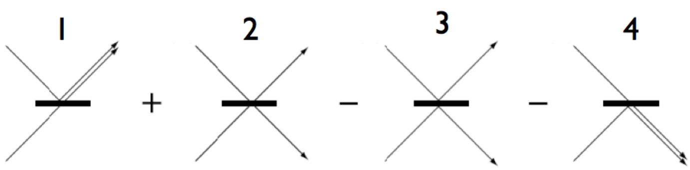



Figure 6: The four scattering amplitudes contributing to the two-photon output.

Crucially, for bosons the operators commute: $\hat{a}_3^\dagger \hat{a}_4^\dagger = \hat{a}_4^\dagger \hat{a}_3^\dagger$. Outcomes 2 and 3 therefore contribute amplitudes $+1/2$ and $-1/2$ to the final state $|1,1\rangle$ and cancel completely. What remains is:

$$\hat{a}_1^\dagger \hat{a}_2^\dagger |0,0\rangle = \frac{1}{2}(\hat{a}_3^{\dagger 2} - \hat{a}_4^{\dagger 2})\,|0,0\rangle = \frac{1}{\sqrt{2}}(|2,0\rangle - |0,2\rangle) \tag{20}$$

Both photons always exit the same port: the probability of a coincidence detection, one photon at $\hat{a}_3$ and one at $\hat{a}_4$, is exactly zero. This is the HOM dip.

#### 4.1.2 Physical meaning

Let us briefly discuss the physical intuition behind the HOM effect. This cancellation is a manifestation of two-photon quantum interference. For identical photons there is no way to determine which photon was reflected and which was transmitted: outcomes 2 and 3 are indistinguishable as they both lead to the same final state $|1,1\rangle$. Their probability amplitudes carry opposite signs and must be added coherently: they interfere destructively.
By contrast, outcomes 1 ($|2,0\rangle$) and 4 ($|0,2\rangle$) each arise from a single unambiguous path with no partner to interfere with. They survive and each occur with probability $1/2$: the photons always bunch into the same output port.
Note that this picture breaks down as soon as the photons become distinguishable. Any difference in an internal degree of freedom (frequency, temporal profile...) makes the path information accessible in principle, the coincidence paths are no longer identical, and the destructive interference is partial or absent. The coincidence probability then rises towards the classical value of $1/2$, and the visibility of the HOM dip becomes a direct quantitative measure of photon indistinguishability.

#### 4.1.3 Simple simulation

To illustrate the HOM effect, we model each photon as a Gaussian wavepacket of width $\sigma$. The coincidence probability as a function of delay $\tau$ is:

$$P_{\text{coinc}}(\tau) = \frac{1}{2}\Big(1 - \big|\langle\psi_1|\psi_2(\tau)\rangle\big|^2\Big) \tag{21}$$

Figure 7 shows the plot from the simulation.

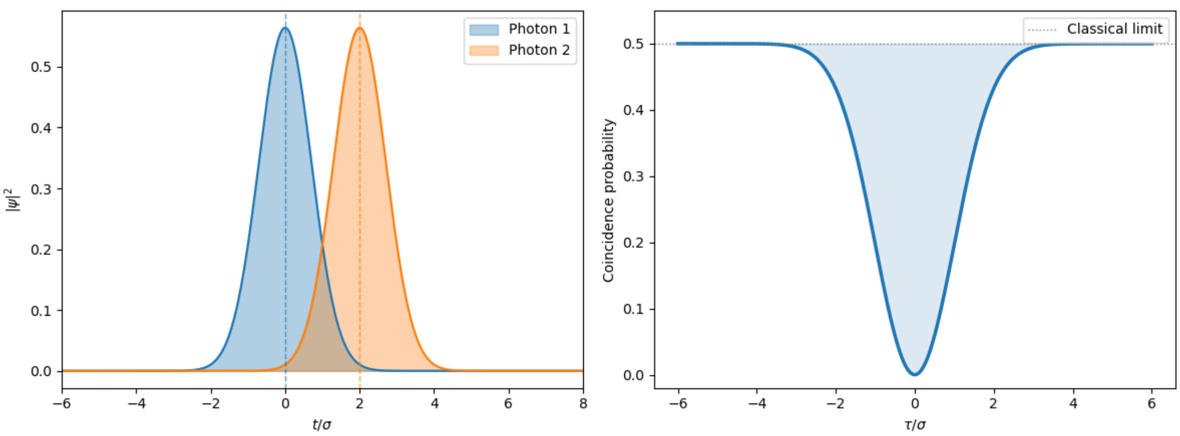



Figure 7: Left: two Gaussian input wavepackets at a relative delay $\tau = 2\sigma$. Right: coincidence probability for identical photons as a function of time delay.

Note that this is a very simplified simulation. We assume both photons have the same width $\sigma$ and neglect several effects present in a real experiment, such polarisation mismatch, chirp, and

multi-photon contamination, all of which would partially fill the dip.
As an illustration of how photon parameters affect the coincidence probability, Figure 8 shows the effect of a frequency mismatch $\Delta\omega$. For Gaussian wavepackets, the overlap evaluates to:

$$\left|\langle\psi_1|\psi_2(\tau)\rangle\right|^2 = e^{-\tau^2/2\sigma^2}\, e^{-\Delta\omega^2\sigma^2/2} \tag{22}$$

so a non-zero $\Delta\omega$ raises the dip minimum from 0 to $\frac{1}{2}(1 - e^{-\Delta\omega^2\sigma^2/2})$.

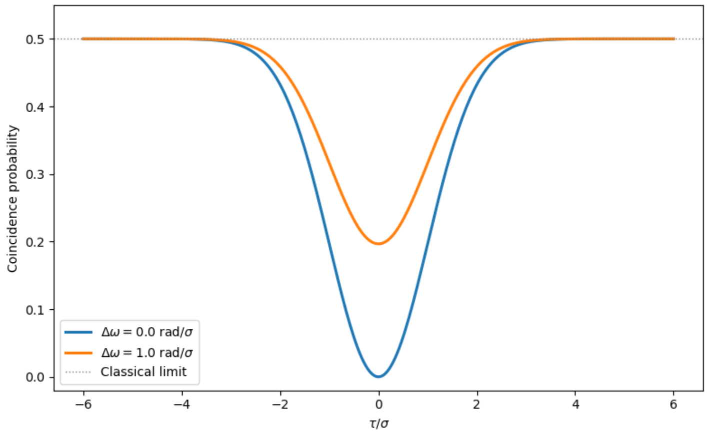



Figure 8: Coincidence probability for two values of frequency mismatch $\Delta\omega$.

As expected, we can see that a non-zero $\Delta\omega$ reduces the wavepacket overlap, partially restoring coincidences and raising the dip minimum above zero.

#### 4.1.4 Deriving an expression for the indistinguishability

We now want to turn the HOM experiment into a quantitative measure. Define the normalised coincidence probability $p_c$ as the probability of detecting one photon in each output port, normalised by the total photon flux in each port:

$$p_c = \frac{\displaystyle\int dt \int d\tau\, g^{(2)}_{34}(t,\tau)}{\left[\displaystyle\int dt\, \langle \hat{a}_3^\dagger(t)\hat{a}_3(t)\rangle\right]\left[\displaystyle\int dt\, \langle \hat{a}_4^\dagger(t)\hat{a}_4(t)\rangle\right]} \tag{23}$$

where $g^{(2)}_{34}(t,\tau) = \langle \hat{a}_3^\dagger(t)\hat{a}_4^\dagger(t+\tau)\hat{a}_3(t)\hat{a}_4(t+\tau)\rangle$ is the cross-correlation between the two output ports. The indistinguishability is then defined as $I = 1 - 2p_c$: it equals 1 when photons always bunch ($p_c = 0$) and 0 for classical light ($p_c = 1/2$).
We now derive $g^{(2)}_{34}$ explicitly. Substituting the beam splitter transformation (eq. 18) into $g^{(2)}_{34}$ and expanding the product of four operators yields 16 terms. Since we work with single-photon inputs, any term containing $\hat{a}_k^2$ or $(\hat{a}_k^\dagger)^2$ vanishes. Only four terms survive [2]:

$$\begin{aligned} g^{(2)}_{34}(t,\tau) = \frac{1}{4}[&\langle \hat{a}_1^\dagger(t)\hat{a}_2^\dagger(t+\tau)\hat{a}_2(t+\tau)\hat{a}_1(t)\rangle \\ &+ \langle \hat{a}_2^\dagger(t)\hat{a}_1^\dagger(t+\tau)\hat{a}_1(t+\tau)\hat{a}_2(t)\rangle \\ &- \langle \hat{a}_1^\dagger(t)\hat{a}_2^\dagger(t+\tau)\hat{a}_1(t+\tau)\hat{a}_2(t)\rangle \\ &- \langle \hat{a}_2^\dagger(t)\hat{a}_1^\dagger(t+\tau)\hat{a}_2(t+\tau)\hat{a}_1(t)\rangle] \end{aligned} \tag{24}$$

We now apply two assumptions.
First, identical statistics means both photons are drawn from the same source and have the

same single-photon correlation functions: $\langle \hat{a}_1^\dagger(t)\hat{a}_1(t')\rangle = \langle \hat{a}_2^\dagger(t)\hat{a}_2(t')\rangle \equiv \langle \hat{a}^\dagger(t)\hat{a}(t')\rangle$. In practice this means both photons have the same spectrum, linewidth, and temporal envelope (they are statistically identical copies).
Second, no phase correlations means the two successive emissions are independent events: there is no fixed phase relationship between the field of photon 1 and the field of photon 2. Mathematically, any expectation value mixing operators from ports 1 and 2 factorises: $\langle \hat{a}_1^\dagger(\cdots)\hat{a}_2(\cdots)\rangle = \langle \hat{a}_1^\dagger(\cdots)\rangle\langle \hat{a}_2(\cdots)\rangle = 0$, since single-photon fields have zero mean. This is what allows the four-operator mixed expectation values to split into products of two-operator ones.
Under these two assumptions the first term becomes:

$$\langle \hat{a}_1^\dagger(t)\hat{a}_2^\dagger(t+\tau)\hat{a}_2(t+\tau)\hat{a}_1(t)\rangle = \langle \hat{a}^\dagger(t)\hat{a}(t)\rangle \, \langle \hat{a}^\dagger(t+\tau)\hat{a}(t+\tau)\rangle$$

and the third term is:

$$\langle \hat{a}_1^\dagger(t)\hat{a}_2^\dagger(t+\tau)\hat{a}_1(t+\tau)\hat{a}_2(t)\rangle = \langle \hat{a}_1^\dagger(t)\hat{a}_1(t+\tau)\rangle\langle \hat{a}_2^\dagger(t+\tau)\hat{a}_2(t)\rangle = \left|\langle \hat{a}^\dagger(t+\tau)\hat{a}(t)\rangle\right|^2$$

The two first terms are equal, and so are the two last. The factor of $\frac{1}{4}$ combines with the contributions and gives an overall $\frac{1}{2}$, and eq. (24) becomes:

$$g_{34}^{(2)}(t,\tau) = \frac{1}{2}\Big[\langle \hat{a}^\dagger(t)\hat{a}(t)\rangle \, \langle \hat{a}^\dagger(t+\tau)\hat{a}(t+\tau)\rangle - \left|\langle \hat{a}^\dagger(t+\tau)\hat{a}(t)\rangle\right|^2\Big] \tag{25}$$

Substituting into $p_c$ and using $I = 1 - 2p_c$, the classical contributions (first term) cancel and one obtains:

$$\boxed{I = \frac{\displaystyle\int_0^\infty dt \int_0^\infty d\tau \left|\langle \hat{a}^\dagger(t+\tau)\hat{a}(t)\rangle\right|^2}{\displaystyle\int_0^\infty dt \int_0^\infty d\tau \, \langle \hat{a}^\dagger(t)\hat{a}(t)\rangle\langle \hat{a}^\dagger(t+\tau)\hat{a}(t+\tau)\rangle}} \tag{26}$$

## 4.2 The approach

Following the failure of the numerical approach to compute the efficiency, it is preferable to avoid computing $I$ by naively evaluating the integrals numerically via quadrature. Even if it worked, this would be slow and subject to discretisation error.
Instead, this section is dedicated to solving it analytically, which is more involved that the efficiency but thankfully possible. Since the full derivation is quite long, we first give the conceptual approach to solving this.
The key insight is that the correlators in $I$ can be written as sums of exponentials in $t$ and $\tau$, which makes the double integrals tractable in closed form. The strategy to get there has three steps:

1. **Density matrix.** We solve the Lindblad master equation analytically via eigendecomposition of the Liouvillian. This gives $rho(t)$, useful for computing both the single-time correlators $\langle \hat{a}^\dagger(t)\hat{a}(t)\rangle$ and the two-time correlator $\langle \hat{a}^\dagger(t+\tau)\hat{a}(t)\rangle$.
2. **Green's function.** The two-time correlator $\langle \hat{a}^\dagger(t+\tau)\hat{a}(t)\rangle$ in the numerator requires knowing how the field propagates forward in time by $\tau$. This is given by the retarded Green's function.
3. **Integration.** Once both quantities are written as sums of exponentials, the double integral over $t$ and $\tau$ reduces to a finite sum of terms of the form $\int_0^\infty e^{\lambda t}\, dt = -1/\lambda$, giving an exact closed-form result for $I$.

## 4.3 Density matrix

Our starting point is once again the Lindblad master equation.

### 4.3.1 Vectorisation formalism

To solve it analytically, we once again use the vectorisation formalism.
Note however that we now work in the 2D basis $\{|e,0\rangle, |g,1\rangle\}$. Unlike the efficiency calculation, we do not need to track the state $|g,0\rangle$ here. Physically, this is because once all population has decayed to the ground state with an empty cavity, it contributes nothing further to the cavity field correlations we measure.
In the Hilbert space, the density matrix is $2\times 2$:

$$\rho = \begin{pmatrix} \rho_{ee} & \rho_{ec} \\ \rho_{ce} & \rho_{cc} \end{pmatrix}$$

The vectorisation procedure flattens this into a column vector.

$$|\rho\rangle\rangle = \begin{pmatrix} \rho_{ee} \\ \rho_{cc} \\ \mathrm{Re}(\rho_{ec}) \\ \mathrm{Im}(\rho_{ec}) \end{pmatrix} \in \mathbb{R}^4$$

We note here that since $\rho_{ce} = \rho_{ec}^*$ (hermiticity), the two off-diagonal elements are not linearly independent. To keep our matrices real (which is numerically cleaner), we decompose the complex coherence into its real and imaginary parts.
The Lindblad equation in vectorised form becomes:

$$\frac{d}{dt}|\rho\rangle\rangle = \hat{L}\,|\rho\rangle\rangle$$

where $\hat{L}$ is the $4\times 4$ Liouvillian superoperator matrix. The formal solution is:

$$|\rho(t)\rangle\rangle = e^{\hat{L}t}\,|\rho(0)\rangle\rangle$$

with the initial condition $|\rho(0)\rangle\rangle = (1,0,0,0)^T$, corresponding to the pure state $|e,0\rangle\langle e,0|$.

### 4.3.2 Eigendecomposition

To evaluate the matrix exponential $e^{\hat{L}t}$ analytically, we use eigendecomposition. If $\hat{L} = V\Lambda V^{-1}$ where $V$ contains the eigenvectors and $\Lambda$ is the diagonal matrix of eigenvalues, then:

$$e^{\hat{L}t} = V e^{\Lambda t} V^{-1} \tag{27}$$

The diagonal exponential is trivial:

$$e^{\Lambda t} = \begin{pmatrix} e^{\lambda_1 t} & 0 & 0 & 0 \\ 0 & e^{\lambda_2 t} & 0 & 0 \\ 0 & 0 & e^{\lambda_3 t} & 0 \\ 0 & 0 & 0 & e^{\lambda_4 t} \end{pmatrix}$$

The product $Ve^{\Lambda t}V^{-1}$ can be written explicitly as a spectral decomposition. Since the diagonal exponential decomposes $e^{\Lambda t} = \sum_i e^{\lambda_i t}\,|e_i\rangle\rangle\langle\langle e_i|$ in the eigenbasis, we obtain:

$$\begin{aligned} e^{\hat{L}t} &= V e^{\Lambda t} V^{-1} \\ &= \sum_i e^{\lambda_i t}\,(V\,|e_i\rangle\rangle)(\langle\langle e_i|\,V^{-1}) \\ &= \sum_i e^{\lambda_i t}\,|v_i\rangle\rangle\langle\langle w_i| \end{aligned} \tag{28}$$

and so it follows that

$$|\rho(t)\rangle\rangle = \sum_{i=1}^{4} e^{\lambda_i t} |v_i\rangle\rangle \langle\langle w_i|\rho(0)\rangle\rangle = \sum_{i=1}^{4} e^{\lambda_i t} \xi_i |v_i\rangle\rangle \tag{29}$$

where $|v_i\rangle\rangle$ are the right eigenvectors of $\hat{L}$ (columns of $V$), $\langle\langle w_i|$ the left eigenvectors of $\hat{L}$ (rows of $V^{-1}$) and $\xi_i$ the overlap. The density matrix thus evolves as a superposition of four independent decay channels, each with rate $\lambda_i$. From the evolved vectorised density matrix, we extract the physical density matrix elements:

$$\rho_{ee}(t),\ \rho_{cc}(t),\ \rho_{ec}(t) = \mathrm{Re}(\rho_{ec}(t)) + i \cdot \mathrm{Im}(\rho_{ec}(t)) \tag{30}$$

These density matrix elements are essential for the next step. The cavity population $\rho_{cc}(t)$ directly gives the single-time intensity correlator $\langle \hat{a}^\dagger(t)\hat{a}(t)\rangle = \rho_{cc}(t)$, which is needed for the denominator of the indistinguishability integral. More importantly, both $\rho_{cc}(t)$ and $\rho_{ec}(t)$ will be combined with the Green's function to compute the two-time correlators $\langle \hat{a}^\dagger(t+\tau)\hat{a}(t)\rangle$.

## 4.4 Green's Function

To compute the two-time correlators, we need to understand how the cavity field evolves across a time delay. This is where the retarded Green's function comes in.
Conceptually, $\hat{G}^R(\tau)$ is a single-particle propagator: it describes how a quantum amplitude evolves forward in time by $\tau$. Combined with the density matrix at time $t$ via the Kadanoff-Baym equation, it will give us access to the two-time field correlations $\langle \hat{a}^\dagger(t+\tau)\hat{a}(t)\rangle$.

### 4.4.1 Green function definition

In the frequency domain, the retarded Green's function satisfies the Dyson equation [3]:

$$(\omega - H - \Sigma^R)\hat{G}^R(\omega) = I \tag{31}$$

where $H$ is the Hamiltonian and $\Sigma^R$ encodes the dissipative self-energies (decay rates). Inverting gives [2]:

$$\hat{G}^R(\omega) = \begin{pmatrix} \omega + i\gamma/2 + i\gamma^*/2 & g \\ g & \omega - \delta + i\kappa/2 \end{pmatrix}^{-1} \tag{32}$$

However, we are interested in Green's function in the time domain. It satisfies the equation of motion [2] (for $\tau > 0$):

$$\begin{aligned} i\frac{\partial \hat{G}^R(\tau)}{\partial \tau} &= i\delta(\tau)\hat{I} + [H + \Sigma^R(0)]\hat{G}^R(\tau) \\ &= [H + \Sigma^R(0)]\hat{G}^R(\tau) \end{aligned} \tag{33}$$

Define the propagation matrix:

$$M = -i[H + \Sigma^R(0)] = \begin{pmatrix} -(\gamma+\gamma^*)/2 & -ig \\ -ig & -i\delta - \kappa/2 \end{pmatrix} \tag{34}$$

Then the equation of motion becomes $\partial\hat{G}^R(\tau)/\partial\tau = M\hat{G}^R(\tau)$, which is solved by:

$$\hat{G}^R(\tau) = e^{M\tau} \tag{35}$$

with the boundary condition $\hat{G}^R(0^+) = I$ because the field at time $t$ is fully correlated with itself.

### 4.4.2 Eigendecomposition

This is exactly the same form as the density matrix evolution, we can evaluate $e^{M\tau}$ analytically using eigendecomposition. Diagonalizing the $2 \times 2$ propagation matrix $M = U\tilde{\Lambda}U^{-1}$, where $\tilde{\Lambda}$ contains the eigenvalues $\mu_i$:

$$\hat{G}^R(\tau) = \begin{pmatrix} G^R_{ee}(\tau) & G^R_{ec}(\tau) \\ G^R_{ce}(\tau) & G^R_{cc}(\tau) \end{pmatrix} = e^{M\tau} = Ue^{\tilde{\Lambda}\tau}U^{-1} = \sum_{j=1}^{2} e^{\mu_j \tau} |v_j\rangle \langle w_j| \tag{36}$$

where $|v_j\rangle$ are the right eigenvectors and $\langle w_j|$ are the left eigenvectors of $M$.
Similarly to the density matrix, we managed to express Green's function as a sum of exponentials.

## 4.5 Two-time correlator

Armed with these results, we can now finally compute the two-time correlator.

### 4.5.1 Kadanoff-Baym relation

The connection between the density matrix at time $t$, the retarded Green function and the field correlation is given by the Kadanoff-Baym equations [3] from non-equilibrium Green's function theory.
This relation reads:

$$\hat{G}^<(t, t+\tau) = \hat{G}^R(\tau)\hat{\rho}(t) \tag{37}$$

where $\hat{G}^<$ is the lesser Green's function encoding the two-time correlations [2]:

$$\hat{G}^<(t, t+\tau) = \begin{pmatrix} \langle \hat{\sigma}^\dagger(t+\tau)\hat{\sigma}(t)\rangle & \langle \hat{\sigma}^\dagger(t+\tau)\hat{a}(t)\rangle \\ \langle \hat{a}^\dagger(t+\tau)\hat{\sigma}(t)\rangle & \langle \hat{a}^\dagger(t+\tau)\hat{a}(t)\rangle \end{pmatrix} \tag{38}$$

From this, the two-time correlator of the cavity field can be extracted by taking the cavity-cavity component of the matrix:

$$\begin{aligned} \langle \hat{a}^\dagger(t+\tau)\hat{a}(t)\rangle &= \langle g, 1| \hat{G}^R(\tau)\hat{\rho}(t) |g, 1\rangle \\ &= G^R_{cc}(\tau)\rho_{cc}(t) + G^R_{ce}(\tau)\rho_{ec}(t) \end{aligned} \tag{39}$$

### 4.5.2 Notes about the components

Notice that having the vector $|\rho(t)\rangle\rangle$ or the matrix $G^R(\tau)$ as sum of exponential means that we are also able to express each individual elements as a sum.
For the sake of completeness, let us express the quantities of interest given in eq. (39).
To start with $|\rho(t)\rangle\rangle$, we go back to equation (29) and can write:

$$\begin{aligned} \rho_{cc}(t) &= \sum_i \xi_i \, [v_i]_2 \, e^{\lambda_i t} \\ \rho_{ec}(t) &= \sum_i \xi_i \left( [v_i]_3 + i[v_i]_4 \right) e^{\lambda_i t} \end{aligned} \tag{40}$$

where $[v_i]_k$ denotes the $k$-th component of $|v_i\rangle\rangle$. Similarly, for $\hat{G}^R$: since $\hat{G}^R(\tau) = \sum_j e^{\mu_j \tau} |v_j\rangle \langle w_j|$, each matrix element reads:

$$\begin{aligned} G^R_{cc}(\tau) &= \sum_j [v_j]_2 [w_j]_2 e^{\mu_j \tau} \\ G^R_{ce}(\tau) &= \sum_j [v_j]_2 [w_j]_1 e^{\mu_j \tau} \end{aligned} \tag{41}$$

where $[v_j]_k$ and $[w_j]_k$ denote the $k$-th components of the right and left eigenvectors of $M$.
The sum in equation (39) can therefore be rewritten:

$$\boxed{\langle \hat{a}^\dagger(t+\tau)\hat{a}(t)\rangle = \sum_{i,j} \alpha_{ij} e^{\lambda_i t + \mu_j \tau}} \tag{42}$$

where $\alpha_{ij}$ collects the scalar prefactors from individual eigendecompositions. Note that these can be complex valued here.

## 4.6 Integration

Finally, we can use our correlators written in this new form (as sums of exponentials) to evaluate their integrals by exploiting the closed form:

$$\int_0^\infty e^{\lambda t} dt = -\frac{1}{\lambda} \tag{43}$$

### 4.6.1 Validity of the closed form

Note that equation (43) is valid under the condition that $\mathrm{Re}(\lambda) < 0$.
We know the Liouvillian describe a dissipative system, where the eigenvalues describe the evolution of eigenmodes, that need to decay to not violate energy conservation.
Similarly, the retarded Green's function describes the propagation of the field's correlation, that decay to 0 when $\tau$ goes to infinity. The eigenvalues therefore have negative real parts. We can also notice that the propagation matrix $M$ is constructed with dissipative terms on the diagonal. We can therefore apply the closed form without worry.

### 4.6.2 Numerator

The numerator of the indistinguishability is the two-time correlator and follows from (42):

$$\begin{aligned}
\int_0^\infty dt \int_0^\infty d\tau \, |\langle \hat{a}^\dagger(t+\tau)\hat{a}(t)\rangle|^2 &= \int_0^\infty dt \int_0^\infty d\tau \left| \sum_{i,j} \alpha_{ij} e^{\lambda_i t + \mu_j \tau} \right|^2 \\
&= \sum_{i,j,k,l} \alpha_{ij}^* \alpha_{kl} \int_0^\infty e^{(\lambda_i + \lambda_k^*)t} dt \int_0^\infty e^{(\mu_j + \mu_l^*)\tau} d\tau \\
&= \sum_{i,j,k,l} \frac{\alpha_{ij}^* \alpha_{kl}}{(\lambda_i + \lambda_k^*)(\mu_j + \mu_l^*)}
\end{aligned} \tag{44}$$

### 4.6.3 Denominator

The denominator is simpler because it involves only single-time correlators. The two factors $\langle \hat{a}^\dagger(t)\hat{a}(t)\rangle$ and $\langle \hat{a}^\dagger(t+\tau)\hat{a}(t+\tau)\rangle$ are both just the instantaneous cavity population at a given time, which is directly $\rho_{cc}$ evaluated at that time:

$$\langle \hat{a}^\dagger(t)\hat{a}(t)\rangle = \rho_{cc}(t) = \sum_i \xi_i \, [v_i]_2 \, e^{\lambda_i t} \tag{45}$$

The denominator integral therefore factorises into a product of two independent integrals:

$$\begin{aligned}
\int_0^\infty dt \int_0^\infty d\tau \, \langle \hat{a}^\dagger(t)\hat{a}(t)\rangle \langle \hat{a}^\dagger(t+\tau)\hat{a}(t+\tau)\rangle &= \int_0^\infty \rho_{cc}(t) \, dt \int_0^\infty \rho_{cc}(\tau) \, d\tau \\
&= \left( \int_0^\infty \sum_i \xi_i \, [v_i]_2 \, e^{\lambda_i t} \, dt \right)^2 \\
&= \left( \sum_i \frac{\xi_i \, [v_i]_2}{\lambda_i} \right)^2
\end{aligned} \tag{46}$$

### 4.7 Plot

This derivation finally allows us to bypass the time integration. Figure 9 shows the indistinguishability computed this way.
Note that from a computational aspect, this method is both exact and efficient across the parameter space (the simulation time is constant regardless of the values of $\kappa$ or $g$).

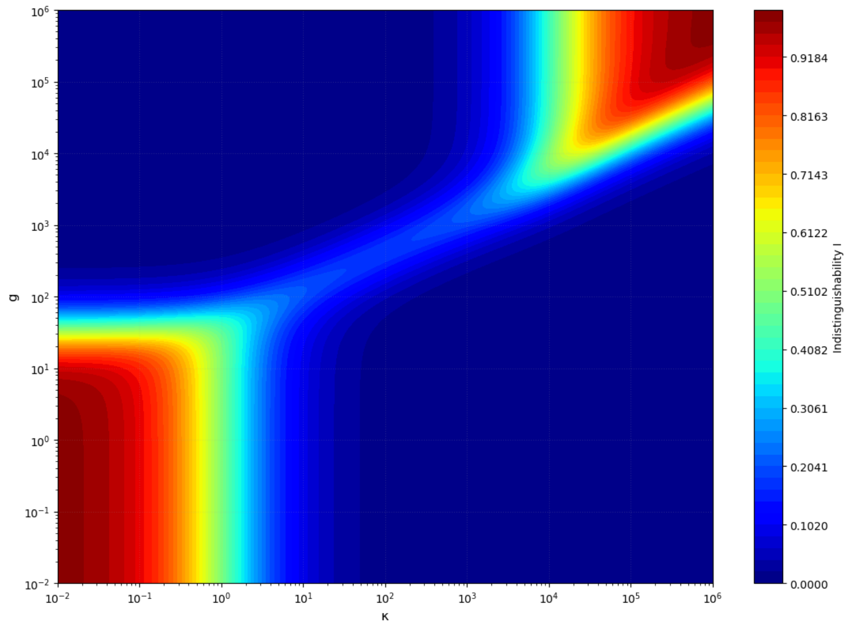



Figure 9: Indistinguishability $I$ as a function of the coupling $g$ and cavity decay rate $\kappa$, for $\gamma^*/\gamma = 10^4$.

## 5 Discussion on the different regimes

Let us now take a step back, and remember our initial goal. The reason we were looking for the efficiency and indistinguishability of these emitter-cavity systems was to find a new regime that isn't currently exploited in single photon emitters designs. Let us try to get an understanding of the maps of $\beta$ (Fig. 3) and $I$ (Fig. 9) physically, and identifying the distinct regions they exhibit.

### 5.1 The cavity rate

Before reading the maps, we introduce the cavity-funneled rate $\Gamma_{\text{cav}}$. It is the emission rate enhanced by the Purcell factor $\Gamma_{\text{cav}} = \gamma F_P$, or how much the cavity speeds up the emission rate relative to free space.
This rate is given by [4]:

$$\Gamma_{\text{cav}} \propto \frac{g^2}{\kappa + \gamma + \gamma^*} \tag{47}$$

Note that this isn't a strict equality, because there is usually a prefactor, that depends on convention. Here, we are doing a qualitative discussion, so we do not worry about it and accept slight quantitative deviation in our considerations.

## 5.2 The efficiency map

At first approximation, the efficiency is ratio between the good and bad emission channels:

$$\beta \approx \frac{\Gamma_{\text{cav}}}{\Gamma_{\text{cav}} + \gamma} = \frac{F_P}{F_P + 1} \tag{48}$$

so that as we can expect, $\beta$ goes to 1 when $F_p = \Gamma_{\text{cav}}/\gamma \gg 1$.
From (47), this condition becomes

$$\frac{g^2}{\gamma(\kappa + \gamma + \gamma^*)} \gg 1 \tag{49}$$

This condition traces the boundary of the high $\beta$ region seen in Fig. 3: First, if $\kappa \gg \gamma^*$ the denominator is dominated by $\kappa$ and the condition becomes $g^2 > \kappa\gamma$, or $g/\gamma > \sqrt{\kappa/\gamma}$. On the log-log $(\kappa/\gamma,\, g/\gamma)$ plane, this is a straight line of slope 1/2.
Then, when $\kappa \ll \gamma^*$ the denominator is dominated by $\gamma^*$ and the condition reduces to $g^2 > \gamma\gamma^*$, or $g/\gamma > \sqrt{\gamma^*/\gamma}$, independent of $\kappa$. This is the horizontal floor of the high $\beta$ region.
We can also note that intuitively, we need $\kappa/\gamma > 1$. If the emission rate of the bad channel is larger than that of the good one, the efficiency will be low. This in the plane, this condition is simply a vertical wall.

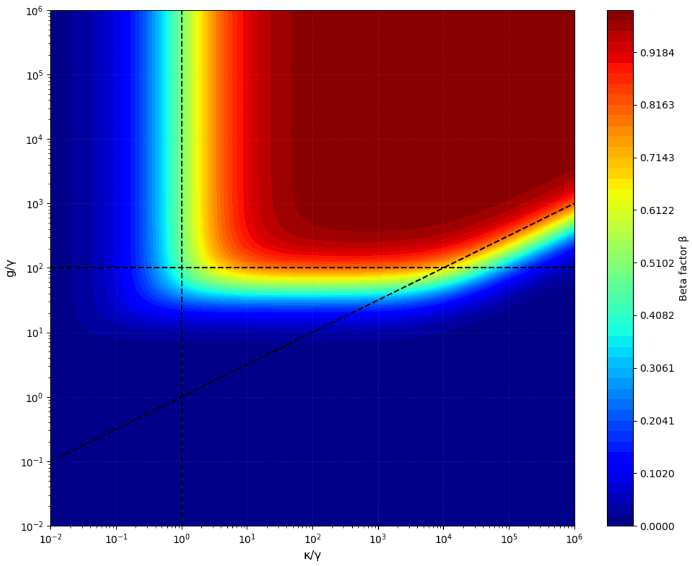



Figure 10: The high-beta region, delimited by the slope 1/2 line, the horizontal line at $\sqrt{\gamma^*/\gamma}$ and the vertical line at $\kappa/\gamma = 1$ for $\gamma^*/\gamma = 10^4$

## 5.3 The indistinguishability map

Unlike the efficiency, the indistinguishability is high in two disconnected regions of Fig. 9.

### 5.3.1 The conventional (Purcell) regime

The natural starting point is the bare emitter. Without a cavity, the emitted photons have indistinguishability

$$I = \frac{\gamma}{\gamma + \gamma^*} \tag{50}$$

set purely by the competition between radiative decay and pure dephasing. For room-temperature solid-state emitters $\gamma^* \gg \gamma$, so $I \approx \gamma/\gamma^* \approx 0$.

The conventional way to recover $I$ is the very same Purcell effect that boosts $\beta$: enhancing the radiative rate $\gamma \to \Gamma_{\text{cav}} = F_P\,\gamma$. The dephasing $\gamma^*$ is unchanged, as it is a property of the emitter and not of the cavity. Replacing $\gamma$ by the cavity-enhanced rate in (50), the cavity-emitted photons now have

$$I \approx \frac{\Gamma_{\text{cav}}}{\Gamma_{\text{cav}} + \gamma^*} = \frac{F_P}{F_P + \gamma^*/\gamma} \tag{51}$$

so indistinguishability is recovered, $I \to 1$, only when

$$F_P \gg \frac{\gamma^*}{\gamma} \Leftrightarrow g^2 \gg \kappa\,\gamma^* \tag{52}$$

This is the high $I$ region in the top-right of Fig. 9. Here $\kappa$ is large, so the cavity is the broadest channel. It is again a slope 1/2 line, but shifted upward by a factor $\sqrt{\gamma^*/\gamma}$. Reaching high $I$ is far more demanding than reaching high $\beta$, since the Purcell factor must now also overcome the dephasing ratio.
Additionally, we also need $\kappa/\gamma > g/\gamma$ to ensure that the cavity acts as a fast drain. Combining with the previous condition, we get the vertical line limiting the conventional regime region:

$$\kappa^2 > g^2 > \kappa\gamma^* \Rightarrow \kappa > \gamma^* \tag{53}$$

This is also where the difficulty of the conventional approach becomes clear, with $\gamma^*/\gamma$ ratios of the order of $10^4$ for room temperature emitters.

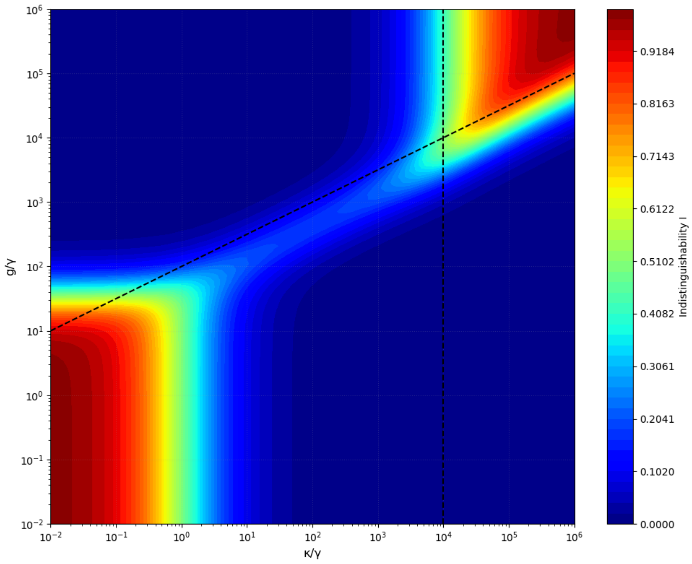



Figure 11: The high-I region, delimited by the slope 1/2 line and the vertical line at $\kappa/\gamma = \gamma^*/\gamma$ for $\gamma^*/\gamma = 10^4$

### 5.3.2 The unconventional (cavity-funneling) regime

The more interesting region is the bottom-left of Fig. 9, where both $\kappa/\gamma$ and $g/\gamma$ are small. Here the efficiency is low, yet the indistinguishability is high, the opposite of the Purcell intuition. This is the cavity-funneling regime of Grange et al. [1].
The mechanism is the following. A high-quality-factor cavity (small $\kappa$, large $Q$) has a spectral linewidth much narrower than the emitter. The weak coupling $g$ transfers only a small fraction of the excitation into the cavity; most of it is lost to direct emission at rate $\gamma$, which is why the efficiency $\beta$ is low. But for the small fraction that is funneled into the cavity, because $g$ is small, there are no Rabi oscillations to carry the photon back to the emitter, and because $\kappa$ is

small, the photon lingers in the cavity and is eventually released through the narrow cavity line. The emitted photon therefore inherits the coherence of the cavity instead of the emitter, and is highly indistinguishable, no matter how large $\gamma^*$ is.
In this sense the cavity acts as a spectral filter, but unlike free-space spectral filtering it funnels rather than discards. Crucially, this regime relies on a high $Q$ (small $\kappa$) rather than a high Purcell factor, and there is therefore hope for this method to be within reach of current fabrication.

### 5.4 The figure of merit

It is unfortunately clear that the unconventional regime gives us a tradeoff, by making the emission of indistinguishable photons possible but at a somewhat low efficiency.
To make this remark quantitative, we follow Grange et al. [1] and introduce the cavity-funneling ratio

$$F = \beta I \tag{54}$$

which is plotted in Fig. 12.

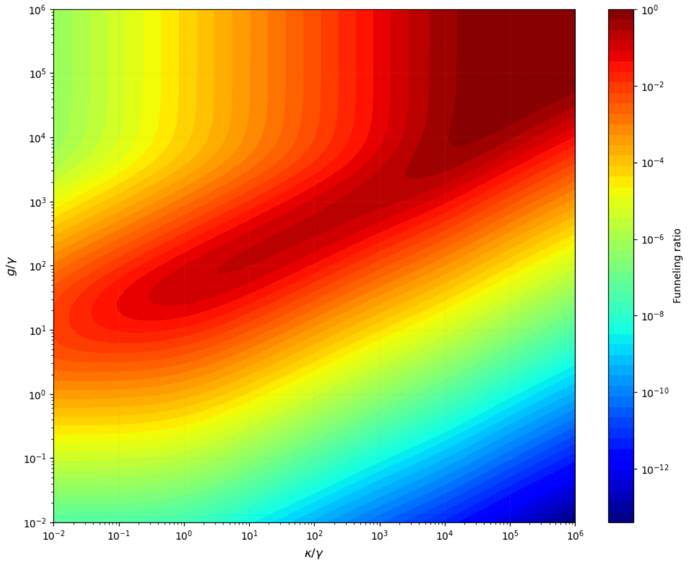



Figure 12: The cavity-funneling ratio for an emitter with $\gamma^*/\gamma = 10^4$.

We can see that a region of reasonable funneling ratio seems accessible.
We now need to evaluate the performance of state of the art cavities in the literature, leading us to our next part.

## 6 Literature examples

Our next goal is to evaluate where current GaN emitter-cavity systems sit in the $(g, \kappa)$ plane. It is worth emphasising that the model derived in Sections 3 and 4 is fully general: $\beta$ and $I$ only depend on the four rates $g$, $\kappa$, $\gamma$, $\gamma^*$, with no further assumption on the material platform or the emitter type. The restriction to GaN here is motivated by the host laboratory's expertise.

### 6.1 Values from state of the art cavities

We surveyed the GaN cavity-QED literature and selected six representative state-of-the-art devices, listed in Table 1. They are from three types of cavities: Fabry-Pérot cavities, photonic crystal cavities and a microring resonator.

| Author | System |
|---|---|
| Schürmann et al. [5] | Fabry-Pérot (micropillar) |
| Hamaguchi et al. [6] | Fabry-Pérot (VCSEL) |
| Niu et al. [7] | PhC cavity (1D nanobeam) |
| Vico Triviño et al. [8] | PhC cavity (L3) |
| Mohamed et al. [9] | PhC cavity (L3) |
| Zeng et al. [10] | Microring resonator |

Table 1: Emitter-cavity systems from the literature.

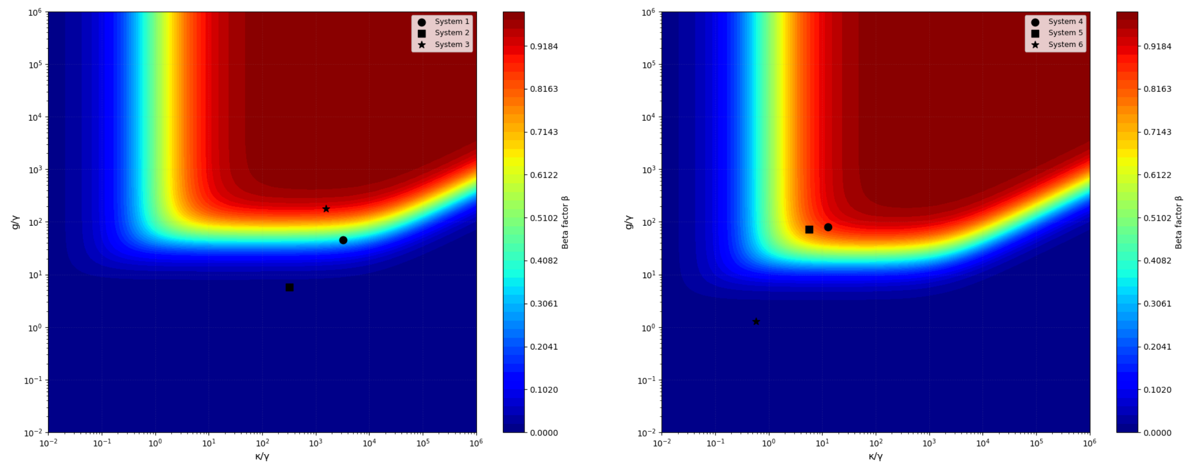
| System | $g/\gamma$ | $\kappa/\gamma$ | $\gamma^*/\gamma$ | $\beta$ | $I$ |
|---|---|---|---|---|---|
| 1 | 45 | 3275 | 12502 | 0.3392 | 0.0007 |
| 2 | 5.8 | 320 | 12502 | 0.0104 | 0.0033 |
| 3 | 180 | 1544 | 12502 | 0.9017 | 0.0080 |
| 4 | 81 | 12.8 | 2090 | 0.8633 | 0.1925 |
| 5 | 72 | 5.7 | 2090 | 0.7834 | 0.1762 |
| 6 | 1.3 | 0.57 | 2090 | 0.0032 | 0.6367 |

Table 2: Parameters and computed figures of merit for the systems in Table 1.

The corresponding figures of merit are reported in Table 2. The methods required to obtain these numbers are given in Appendix B.
The dissipative rates $\gamma$ and $\gamma^*$, on the other hand, are not properties of the cavity but of the emitter itself. For these we use two reference GaN emitters characterised at LASPE and for each cavity pick the emitter with the closest wavelength. The detailed values for each device are also given in Appendix B.

## 6.2 Mapping of state of the art

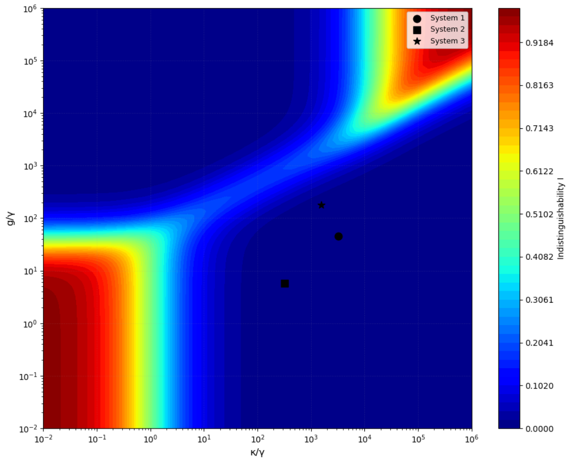
The systems can now be placed on the parameter space. They are separated by emitters, as shown bellow:



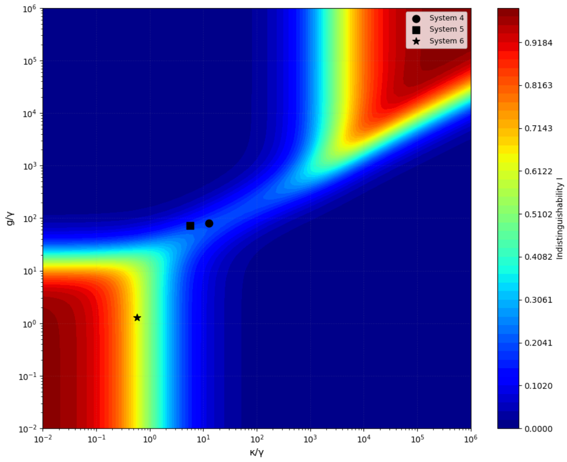

(a) Emitter A

(b) Emitter B

Figure 13: Efficiency $\beta$ in the parameter space for the 6 systems.

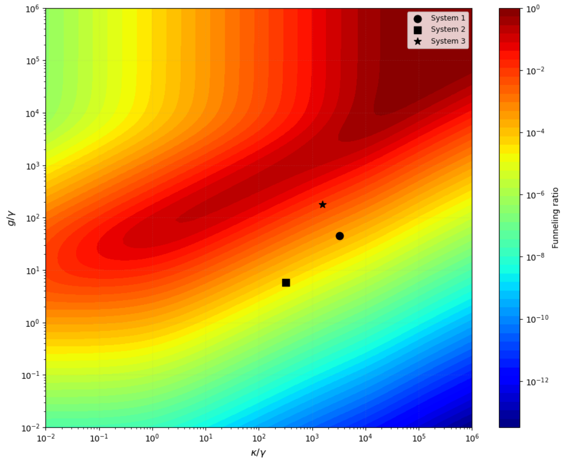



(a) Emitter A

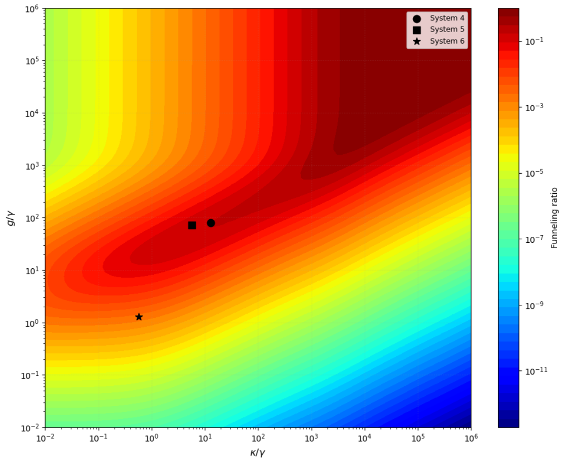



(b) Emitter B

Figure 14: Indistinguishability $I$ in the parameter space for the 6 systems.




(a) Emitter A




(b) Emitter B

Figure 15: Cavity-funneling ratio in the parameter space for the 6 systems.

Across the three figures of merit, the microring resonator (system 6) stands out as the most promising platform. While its efficiency $\beta$ is modest, it is the only device to reach the low-$\kappa$, near-resonant regime where both a high indistinguishability $I$ and a favourable cavity-funneling ratio are achievable, making it the natural candidate to pursue.

# 7 Conclusion

In this project we set out to recover the cavity-funneling model of Grange et al. and apply it to GaN emitter-cavity systems. We first re-derived the efficiency $\beta$ and the indistinguishability $I$ from the Lindblad master equation, both analytically after the naive numerical integration failed due to the stiffness of the problem. We then mapped these two figures of merit across the $(g, \kappa)$ parameter space, identified the conventional and unconventional regimes, and finally placed six state-of-the-art GaN cavities from the literature on these maps.
Across them, the microring resonator stands out as the most promising candidate. Its efficiency is somewhat low, but it is the only device that reaches a low enough $\kappa$, near-resonant regime where both a high indistinguishability and a favourable cavity-funneling ratio become accessible.

This is a direct consequence of the very high quality factors achievable with microrings.
With more time, the natural next step would be a dedicated mathematical modelling of a GaN microring resonator, which we would have liked to carry out but could not within the scope of this project. Such a model would predict the achievable $g$ and $\kappa$ from the ring geometry, and indicate what kind of microring would sit deepest in the funneling regime. Even with all the approximations made here, the estimates suggest that a workable system is not out of reach in GaN, so the cavity-funneling approach seems worth pursuing experimentally, once a proper modelling of the microring has clarified which design to target.

# A Analytical derivation of the efficiency

This appendix completes the element-by-element derivation of the Lindblad equation in the basis $\{|g,0\rangle, |g,1\rangle, |e,0\rangle\}$ that was sketched in Section 3.3. We use the shorthands $g$, $c$, $e$ as indices for the three basis states $|g,0\rangle$, $|g,1\rangle$, $|e,0\rangle$ respectively, e.g. $\rho_{ec} \equiv \langle e,0|\,\rho\,|g,1\rangle$.

## A.1 Commutator contributions

For any element $(i,j)$:

$$\langle i|\,[H,\rho]\,|j\rangle = \langle i|\,H\rho\,|j\rangle - \langle i|\,\rho H\,|j\rangle = \sum_k \langle i|\,H\,|k\rangle\,\rho_{kj} - \sum_k \rho_{ik}\,\langle k|\,H\,|j\rangle$$

With $H = \hbar g(|e,0\rangle\langle g,1| + |g,1\rangle\langle e,0|)$, the only non-vanishing matrix elements of $H$ are:

$$\langle e,0|\,H\,|g,1\rangle = \langle g,1|\,H\,|e,0\rangle = \hbar g.$$

The two sums therefore collapse to at most one term each. Since $H$ acts trivially on $|g,0\rangle$, we have $-\frac{i}{\hbar}[H,\rho]_{gg} = 0$ immediately. The $\dot{\rho}_{ee}$ contribution was already computed in the main text. The two remaining independent elements are $\dot{\rho}_{cc}$ and $\dot{\rho}_{ec}$; the rest follow by hermiticity. For $\dot{\rho}_{cc}$:

$$\begin{aligned}
-\frac{i}{\hbar}\langle g,1|\,[H,\rho]\,|g,1\rangle &= -\frac{i}{\hbar}\Big(\langle g,1|\,H\rho\,|g,1\rangle - \langle g,1|\,\rho H\,|g,1\rangle\Big) \\
&= -\frac{i}{\hbar}\left(\sum_k \langle g,1|\,H\,|k\rangle\,\langle k|\,\rho\,|g,1\rangle - \sum_l \langle g,1|\,\rho\,|l\rangle\,\langle l|\,H\,|g,1\rangle\right) \\
&= -\frac{i}{\hbar}\Big(\hbar g\,\rho_{ec} - \hbar g\,\rho_{ce}\Big) = +ig(\rho_{ce} - \rho_{ec})
\end{aligned}$$

For $\dot{\rho}_{ec}$:

$$\begin{aligned}
-\frac{i}{\hbar}\langle e,0|\,[H,\rho]\,|g,1\rangle &= -\frac{i}{\hbar}\Big(\langle e,0|\,H\rho\,|g,1\rangle - \langle e,0|\,\rho H\,|g,1\rangle\Big) \\
&= -\frac{i}{\hbar}\left(\sum_k \langle e,0|\,H\,|k\rangle\,\langle k|\,\rho\,|g,1\rangle - \sum_l \langle e,0|\,\rho\,|l\rangle\,\langle l|\,H\,|g,1\rangle\right) \\
&= -\frac{i}{\hbar}\Big(\hbar g\,\rho_{cc} - \hbar g\,\rho_{ee}\Big) = +ig(\rho_{ee} - \rho_{cc})
\end{aligned}$$

and $\dot{\rho}_{ce} = \dot{\rho}_{ec}^*$ follows by hermiticity.

## A.2 Dissipator contributions

The same procedure used for $L_1$ in Section 3.3 is now applied to the two remaining collapse operators.

### A.2.1 Cavity decay

First, $L_2^\dagger L_2 = \kappa\,|g,1\rangle\langle g,1| = \kappa P_c$, so the anticommutator term gives:

$$-\frac{\kappa}{2}\langle i|\,\{P_c,\rho\}\,|j\rangle = -\frac{\kappa}{2}(\langle i|\,P_c\rho\,|j\rangle + \langle i|\,\rho P_c\,|j\rangle)$$

And the $L_2\rho L_2^\dagger$ term gives:

$$\kappa\,\langle i|g,0|i|g,0\rangle\,\langle g,1|\,\rho\,|g,1\rangle\,\langle g,0|j|g,0|j\rangle = \kappa\,\delta_{i,g}\delta_{j,g}\,\rho_{cc}$$

So $L_2$ drains $\rho_{cc}$ into $\rho_{gg}$ at rate $\kappa$, and damps every coherence involving $|g,1\rangle$ at rate $\kappa/2$, while leaving $\rho_{ge}$ untouched.

### A.2.2 Pure dephasing

First, $L_3^\dagger L_3 = \gamma^* \left|e,0\right\rangle\!\left\langle e,0\right| = \gamma^* P_e$, so the anticommutator term gives:

$$-\frac{\gamma^*}{2} \langle i| \{P_e, \rho\} |j\rangle = -\frac{\gamma^*}{2} (\langle i| P_e\rho |j\rangle + \langle i| \rho P_e |j\rangle)$$

And the $L_3 \rho L_3^\dagger$ term gives:

$$\gamma^* \langle i|e,0|i|e,0\rangle \langle e,0| \rho |e,0\rangle \langle e,0|j|e,0|j\rangle = \gamma^* \delta_{i,e}\delta_{j,e}\, \rho_{ee}$$

The two contributions on the diagonal element $(e, e)$ exactly cancel ($\gamma^* \rho_{ee} - \gamma^* \rho_{ee} = 0$), confirming that pure dephasing transfers no population. So $L_3$ randomises every coherence involving $|e,0\rangle$ at rate $\gamma^*/2$, but leaves coherences between $|g,0\rangle$ and $|g,1\rangle$ ($\rho_{gc}$) and the populations untouched.

## A.3 Full equations of motion

Summing the commutator contribution and the three dissipators ($L_1$ from Section 3.3, $L_2$, $L_3$) gives the complete set of equations of motion. The populations and the $\rho_{ec}$ coherence reproduce the system used in Section 3.3:

$$\begin{aligned}
\dot\rho_{ee} &= -\gamma\, \rho_{ee} - ig(\rho_{ce} - \rho_{ec}), \\
\dot\rho_{cc} &= -\kappa\, \rho_{cc} + ig(\rho_{ce} - \rho_{ec}), \\
\dot\rho_{gg} &= +\gamma\, \rho_{ee} + \kappa\, \rho_{cc}, \\
\dot\rho_{ec} &= -\tfrac{\gamma+\kappa+\gamma^*}{2}\, \rho_{ec} + ig(\rho_{ee} - \rho_{cc}).
\end{aligned}$$

The remaining coherences decouple from the populations and decay independently:

$$\begin{aligned}
\dot\rho_{gc} &= -\tfrac{\kappa}{2}\, \rho_{gc} + ig\, \rho_{ge}, \\
\dot\rho_{ge} &= -\tfrac{\gamma+\gamma^*}{2}\, \rho_{ge} + ig\, \rho_{gc}.
\end{aligned}$$

Because the initial state $\rho_0 = \left|e,0\right\rangle\!\left\langle e,0\right|$ has $\rho_{gc}(0) = \rho_{ge}(0) = 0$, these last two coherences stay zero throughout the evolution and play no role in the efficiency calculation. This justifies the reduction to the three-variable system $(\rho_{ee}, \rho_{cc}, \operatorname{Im} \rho_{ec})$ used in Section 3.3. The conjugate elements $\rho_{cg}$, $\rho_{eg}$, $\rho_{ce}$ follow trivially by hermiticity.

# B Parameters of the QE-cavity system

The parameters used in our simulations are not always directly reported in the literature. It is therefore useful to know how to convert between measured quantities and the ones relevant here.

## B.1 Cavity decay rate and quality factor

$$\kappa = \frac{\omega_0}{Q} = \frac{2\pi c}{\lambda_0 Q} \tag{55}$$

where:

- $\omega_0$ is the resonance frequency of the cavity
- $\kappa$ is the cavity decay rate (linewidth, in angular frequency units)
- $Q$ is the quality factor
- $c$ is the speed of light
- $\lambda$ is the wavelength

## B.2 Linewidth and dephasing

The spectral linewidth of the emitter is related to the coherence time $T_2$ by:

$$\mathrm{FWHM} = \frac{1}{T_2} = \frac{2\pi c\,\Delta\lambda}{\lambda^2} \tag{56}$$

where $\Delta\lambda$ is the linewidth measured in wavelength units and $\lambda$ is the central wavelength. The total dephasing rate $1/T_2$ has contributions from both radiative decay and pure dephasing:

$$\mathrm{FWHM} = \frac{1}{T_1} + \frac{2}{T_2^*} = \gamma + \gamma^* \tag{57}$$

where:

- $T_2$ is the total coherence time
- $T_1$ is the excited state lifetime (radiative decay), with $\gamma = 1/T_1$
- $T_2^*$ is the pure dephasing time, with $\gamma^* = 2/T_2^*$

## B.3 Coupling factor $g$

The emitter-cavity coupling strength is estimated from cavity and emitter parameters as:

$$g = \sqrt{\frac{3\lambda^2 c}{8\pi n^3 V_c T_1}} \tag{58}$$

where:

- $\lambda$ is the resonant wavelength
- $c$ is the speed of light
- $n$ is the refractive index
- $V_c$ is the cavity mode volume
- $T_1$ is the radiative lifetime of the emitter

## B.4 Mode volume

The cavity mode volume $V_c$ is the parameter that, together with the radiative lifetime $T_1$, sets the magnitude of the coupling strength $g$ via Eq. (58). Unlike the quality factor, $V_c$ is rarely reported explicitly in experimental papers, so it must in most cases be estimated from the cavity geometry or from canonical values associated with each cavity family. We summarise below how $V_c$ was obtained for the cavities considered in this work.

### B.4.1 Photonic crystal cavities

For 2D PhC slab cavities, modal volumes are not reported in either experimental paper from Mohamed et al. [9] and Vico Triviño et al. [8]. We rely instead on canonical values from numerical calculations for L3 [11] and H0 [12] types, confirmed for GaN photonic crystal cavities [13]:

$$V_{\mathrm{L3}} \simeq 0.7\,(\lambda/n)^3$$
$$V_{\mathrm{H0}} \simeq 0.23\,(\lambda/n)^3$$

Using $n_{\mathrm{GaN}} = 2.35$ in the near-IR, this gives:

- L3 at $\lambda = 1300$ nm [8]: $V_c \simeq 0.12\ \mu\text{m}^3$
- L3 at $\lambda = 1545$ nm [9]: $V_c \simeq 0.20\ \mu\text{m}^3$
- H0 at $\lambda = 1513$ nm [9]: $V_c \simeq 0.065\ \mu\text{m}^3$

For 1D PhC nanobeam, the paper from Niu et al. [7] is the only one in our set that explicitly reports a mode volume: $V_c = 1.7\,(\lambda/n)^3$, with $n = 2.5$ for GaN at $\lambda = 454$ nm.
With these numbers,

$$V_c = 1.7 \times \left(\frac{454\ \text{nm}}{2.5}\right)^3 \simeq 1.0 \times 10^{-2}\ \mu\text{m}^3.$$

Despite being larger in normalised units than the L3 or H0 slab values, this is the smallest absolute mode volume in our comparison, reflecting the strong sub-wavelength field confinement characteristic of one-dimensional photonic crystal nanobeam cavities [14].

### B.4.2 Microring resonator

For a microring resonator, the mode volume is set by the propagating waveguide mode multiplied by the ring circumference:

$$V_c \simeq A_{\text{eff}} \cdot 2\pi R$$

where $A_{\text{eff}}$ is the effective transverse mode area. With $R = 60\ \mu$m [10] and $A_{\text{eff}} \simeq 1.5$–$2\ \mu\text{m}^2$ for the GaN ridge waveguide geometry used (2.25 $\mu$m width and 1 $\mu$m thickness [15]).
We obtain

$$V_c \simeq 5\text{–}7 \times 10^2\ \mu\text{m}^3$$

We adopt $V_c \simeq 650\ \mu\text{m}^3$.

### B.4.3 Fabry-Pérot cavities

Unlike photonic crystal cavities, a VCSEL confines light as a Gaussian beam between two DBR mirrors, so it spreads over a spot of radius $w_0$ and extends over the cavity length $L$. The mode volume is therefore set by the beam geometry and estimated [16] as

$$V_c \simeq \frac{\pi w_0^2\, L_{\text{eff}}}{4}$$

where $L_{\text{eff}}$ is the effective cavity length, which includes the geometric cavity length and the field penetration into the DBRs.
Hamaguchi et al. [6] report two device geometries:

- Curved-mirror VCSEL: $L = 27\ \mu$m, $w_0 \approx 0.65\ \mu$m, $V_c \approx 9$–$12\ \mu\text{m}^3$.

The first estimate is consistent with the value $V_c \simeq 12\ \mu\text{m}^3$ reported by Lu et al. [17] for a similar blue GaN VCSEL with hybrid DBRs. We adopt $V_c = 10\ \mu\text{m}^3$.
For the micropillar, another type of F-P cavity, Schürmann et al. [5] report a Purcell factor: $F_P = 3.5$ at $\lambda = 251$ nm, with $Q = 1600$ and a 400 nm-diameter nanopillar containing GaN/AlN quantum dots in an $Al_{0.79}Ga_{0.21}N$.
The Purcell factor relates $Q$ and $V_c$ [18] through:

$$F_P = \frac{3}{4\pi^2}\left(\frac{\lambda}{n}\right)^3 \frac{Q}{V_c}$$

which can be inverted to yield

$$V_c = \frac{3}{4\pi^2 F_P}\left(\frac{\lambda}{n}\right)^3 Q$$

Using $n_{\text{AlGaN}} \simeq 2.2$ at the cavity-mode wavelength, this gives $V_c \simeq 0.05\ \mu\text{m}^3$, approximately $35\,(\lambda/n)^3$. This is significantly larger than the physical pillar volume because the mode penetrates substantially into the DBR stack, as the authors report in their paper (Fig. 10).

## B.5 Unit conventions for coupling and decay rates

The parameters $g$, $\kappa$, $\gamma$, and $\gamma^*$ appear in the literature under several different conventions, which can cause confusion when comparing values across papers.

- **Energy units** ($\hbar g$, $\hbar\kappa$ in eV or $\mu$eV)
- **Angular frequency** ($g$, $\kappa$ in rad/s)
- **Ordinary frequency** ($g/2\pi$, $\kappa/2\pi$ in Hz)

In this report, physical quantities almost always appear as dimensionless ratios ($\kappa/\gamma$, $g/\gamma$, $\gamma^*/\gamma$), so the choice of convention cancels out. When an actual frequency value is needed, we use angular frequencies, written as $2\pi f$ where $f$ is in ordinary frequency.

## B.6 Emitters

For the values of $\gamma$ and $\gamma^*$, we use typical values of emitters characterised at LASPE, matching the one with the closest frequency.

- Emitter A $\lambda = 650\,\text{nm}$, $T_1 = 0.7\,\text{ns}$, FWHM $= 4\,\text{nm}$
- Emitter B $\lambda = 1350\,\text{nm}$, $T_1 = 0.2\,\text{ns}$, FWHM $= 10\,\text{nm}$

Using equations (57) and (56), we find:

- Emitter A: $\gamma = 2\pi \times 227\,\text{MHz}$, $\gamma^* = 2\pi \times 2838\,\text{GHz}$, $\gamma^*/\gamma = 12502$
- Emitter B: $\gamma = 2\pi \times 796\,\text{MHz}$, $\gamma^* = 2\pi \times 1644\,\text{GHz}$, $\gamma^*/\gamma = 2090$

## B.7 Compiling parameters from literature

Finally, using the steps described above, we can summarise the parameters from all the cavities considered in this work in the following Table:

| # | $Q$ | $\lambda_0$ (nm) | $V_c$ (µm$^3$) | $\kappa/\gamma$ | $g/\gamma$ |
|---|---|---|---|---|---|
| 1 | 1600 | 252 | 0.05 | 3275 | 45 |
| 2 | 9078 | 454 | 10 | 320 | 5.8 |
| 3 | 1900 | 450 | 0.01 | 1544 | 180 |
| 4 | 22500 | 1305 | 0.12 | 12.8 | 81 |
| 5 | 44000 | 1512 | 0.2 | 5.7 | 72 |
| 6 | 430000 | 1550 | 650 | 0.57 | 1.3 |

Table 3: Cavity parameters extracted from the literature, used to compute $\kappa$.

# References


[1] Thomas Grange et al. "Cavity-Funneled Generation of Indistinguishable Single Photons from Strongly Dissipative Quantum Emitters". In: *Physical Review Letters* 114 (2015), p. 193601. DOI: 10.1103/PhysRevLett.114.193601.

[2] Thomas Grange et al. *Supplemental Material: Cavity-Funneled Generation of Indistinguishable Single Photons from Strongly Dissipative Quantum Emitters*. Supplemental Material for Phys. Rev. Lett. **114**, 193601 (2015). 2015. DOI: 10.1103/PhysRevLett.114.193601.

[3] Hartmut Haug and Antti-Pekka Jauho. *Quantum Kinetics in Transport and Optics of Semiconductors*. 2nd ed. Vol. 123. Springer Series in Solid-State Sciences. Springer, 2008. ISBN: 978-3-540-73561-8. DOI: 10.1007/978-3-540-73564-9.

[4] T. Hümmer et al. "Weak and strong coupling regimes in plasmonic QED". In: *Physical Review B* 87.11 (2013), p. 115419. DOI: 10.1103/PhysRevB.87.115419.

[5] Hannes Schürmann et al. "GaN Quantum Dots in Resonant Cavity Nanopillars as Deep-UV Single-Photon Sources". In: *physica status solidi (RRL) – Rapid Research Letters* 18 (2024), p. 2400188. DOI: 10.1002/pssr.202400188.

[6] Tatsushi Hamaguchi, Hiroshi Nakajima, and Noriyuki Fuutagawa. "GaN-based Vertical-Cavity Surface-Emitting Lasers Incorporating Dielectric Distributed Bragg Reflectors". In: *Applied Sciences* 9.4 (2019), p. 733. DOI: 10.3390/app9040733.

[7] Nan Niu et al. "Ultra-low threshold gallium nitride photonic crystal nanobeam laser". In: *Applied Physics Letters* 106.23 (2015), p. 231104. DOI: 10.1063/1.4922211.

[8] N. Vico Triviño et al. "Integrated photonics on silicon with wide bandgap GaN semiconductor". In: *Applied Physics Letters* 105.23 (2014), p. 231119. DOI: 10.1063/1.4903861.

[9] M. S. Mohamed et al. "Efficient continuous-wave nonlinear frequency conversion in high-Q gallium nitride photonic crystal cavities on silicon". In: *APL Photonics* 2.3 (2017), p. 031301. DOI: 10.1063/1.4974311.

[10] Hong Zeng et al. "Quantum Light Generation Based on GaN Microring toward Fully On-Chip Source". In: *Physical Review Letters* 132 (2024), p. 133603. DOI: 10.1103/PhysRevLett.132.133603.

[11] Yoshihiro Akahane et al. "High-Q photonic nanocavity in a two-dimensional photonic crystal". In: *Nature* 425 (2003), pp. 944–947. DOI: 10.1038/nature02063.

[12] Zhengbin Zhang and Min Qiu. "Small-volume waveguide-section high-Q microcavities in 2D photonic crystal slabs". In: *Optics Express* 12.17 (2004), pp. 3988–3995. DOI: 10.1364/OPEX.12.003988.

[13] Momchil Minkov and Vincenzo Savona. "Automated optimization of photonic crystal slab cavities". In: *Scientific Reports* 4 (2014), p. 5124. DOI: 10.1038/srep05124.

[14] Qimin Quan and Marko Lončar. "Deterministic design of wavelength scale, ultra-high Q photonic crystal nanobeam cavities". In: *Optics Express* 19.19 (2011), pp. 18529–18542. DOI: 10.1364/OE.19.018529.

[15] Yanzhen Zheng et al. "Integrated Gallium Nitride Nonlinear Photonics". In: *Laser & Photonics Reviews* 16 (2022), p. 2100071. DOI: 10.1002/lpor.202100071.

[16] David Hunger et al. "A fiber Fabry–Perot cavity with high finesse". In: *New Journal of Physics* 12 (2010), p. 065038. DOI: 10.1088/1367-2630/12/6/065038.

[17] Tien-Chang Lu et al. "Development of GaN-Based Vertical-Cavity Surface-Emitting Lasers". In: *IEEE Journal of Selected Topics in Quantum Electronics* 15.3 (2009), pp. 850–860. DOI: 10.1109/JSTQE.2009.2013181.

[18] Kerry J. Vahala. “Optical microcavities”. In: *Nature* 424 (2003), pp. 839–846. DOI: 10.1038/nature01939.